

Theoretical studies of excitonic transitions in two dimensional materials under photoexcitation of scalar and vector vortex beams

1. Introduction
2. Spin-orbital angular momentum interaction in light
3. Light-matter interaction
4. Exciton wave packet

Summary and outlook

1. Introduction

2. Spin-orbital angular momentum interaction in light

3. Light-matter interaction

4. Exciton wave packet

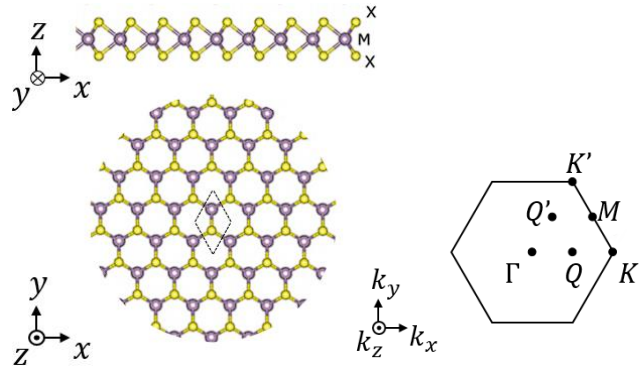
Summary and outlook

# 1. Introduction

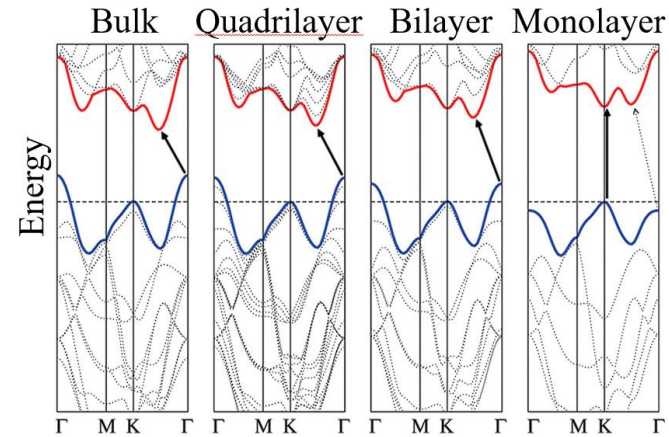
What material ?

# 1. Introduction

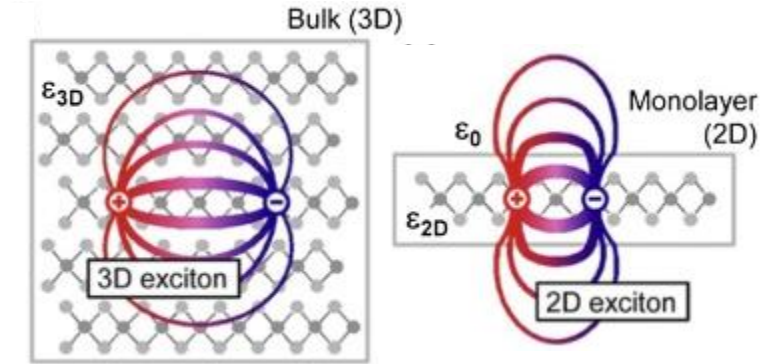
- Two dimensional (2D) Transition metal dichalcogenide (TMD)



- Structures of a typical 2D TMD material  
A. Krasnok & A. Alù, *Appl. Sci.* **8**, 1157 (2018)



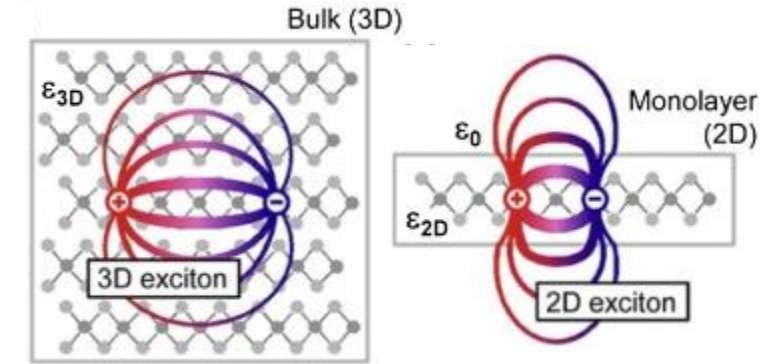
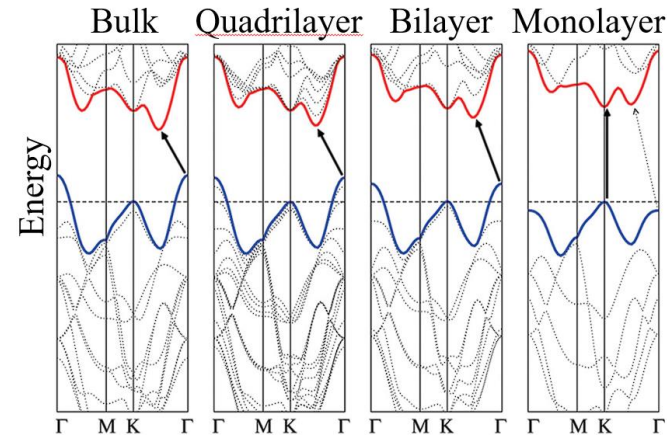
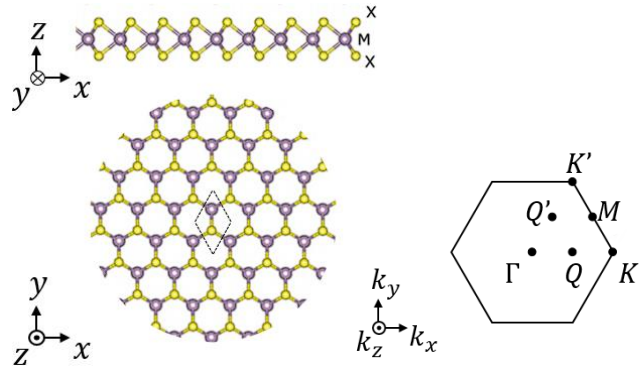
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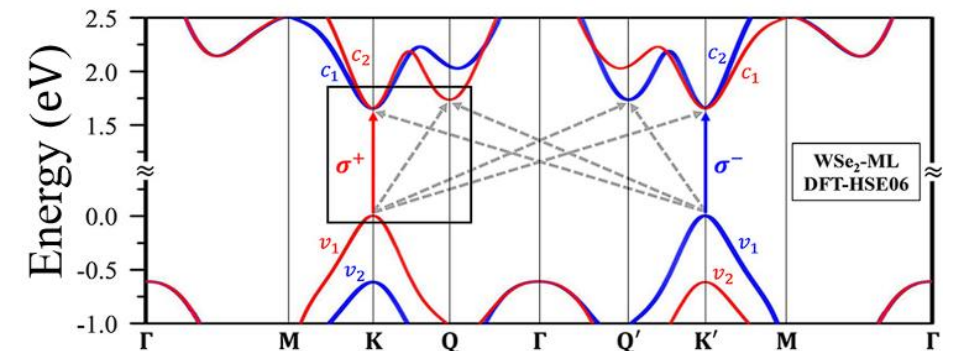
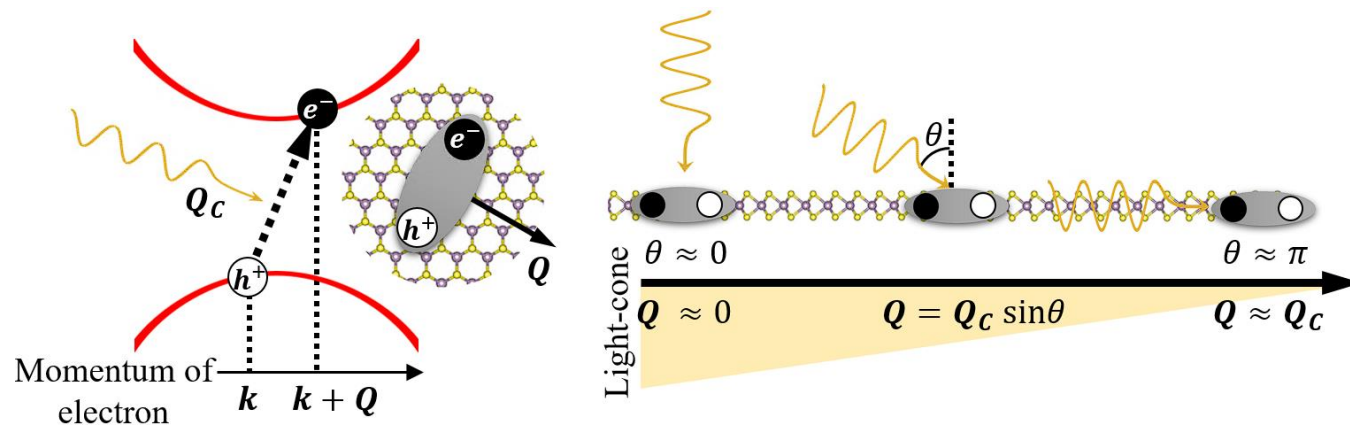


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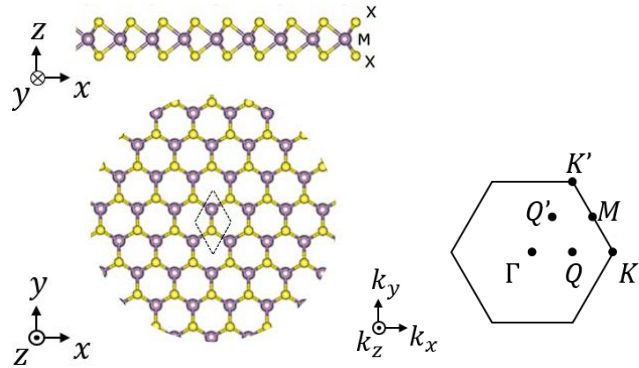
- Exciton in 2D TMD



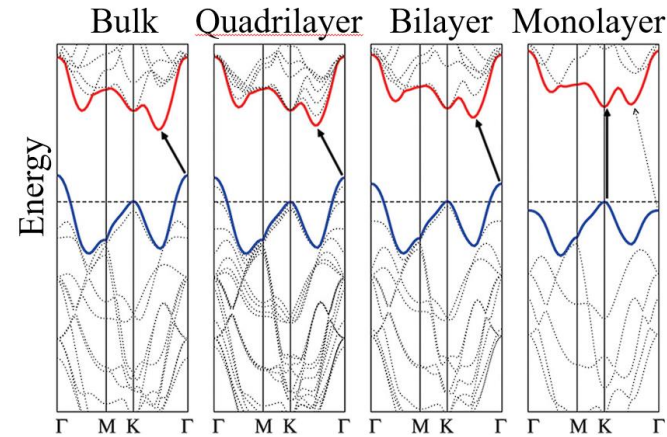
- WSe<sub>2</sub>, calculated band structure  
G.-H. Peng *et al.*, *Nano Lett.* **19**, 2299 (2019)

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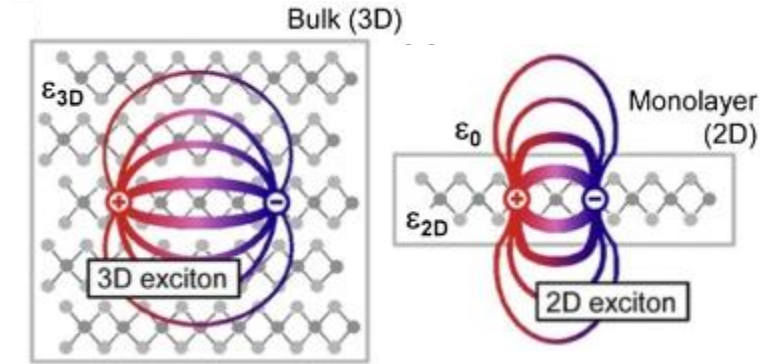
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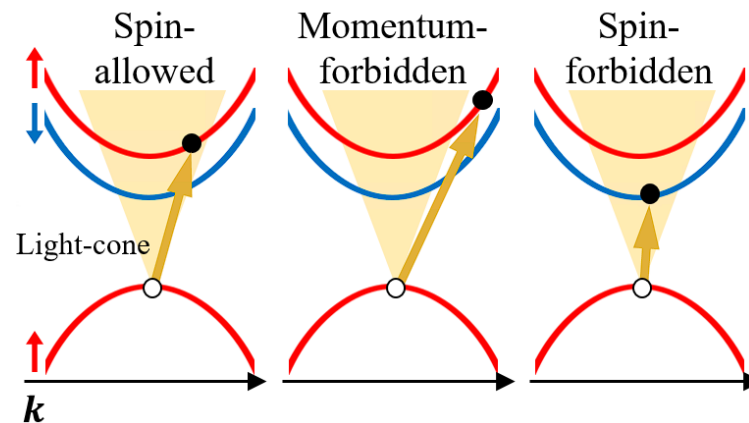


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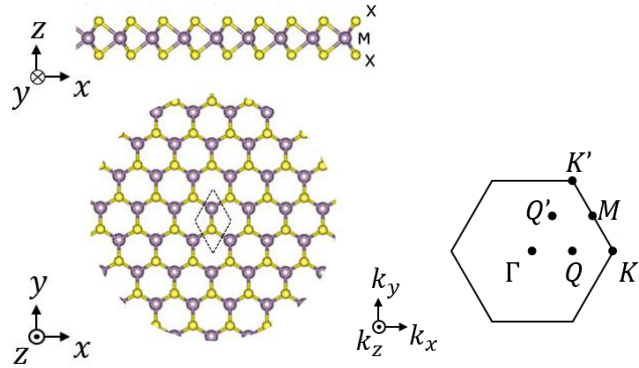
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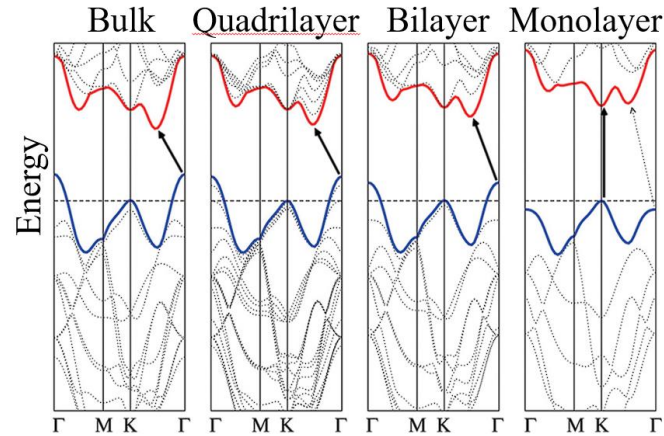
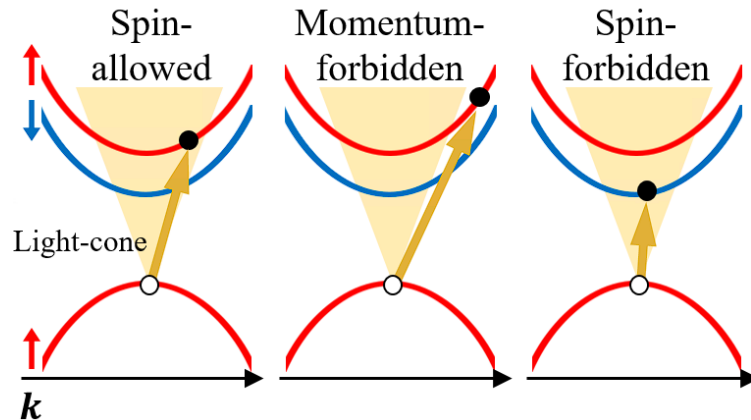
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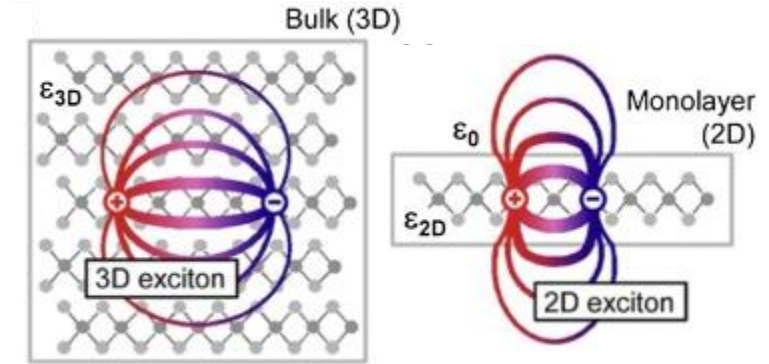
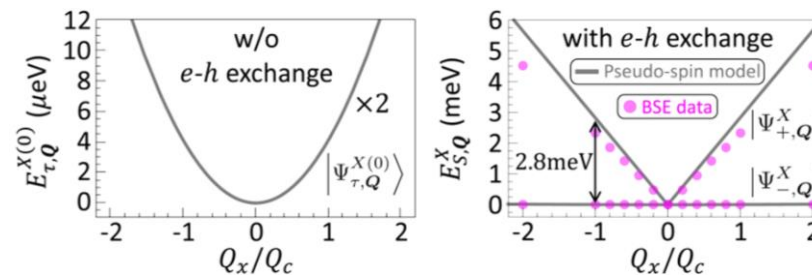
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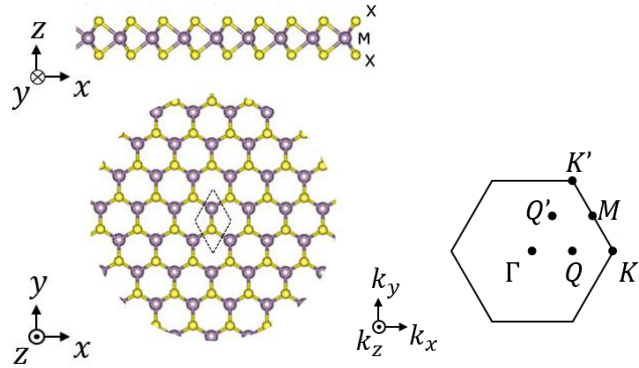
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G.-H. Peng *et al.*, *Phys. Rev. B* **106**, 155304 (2022)



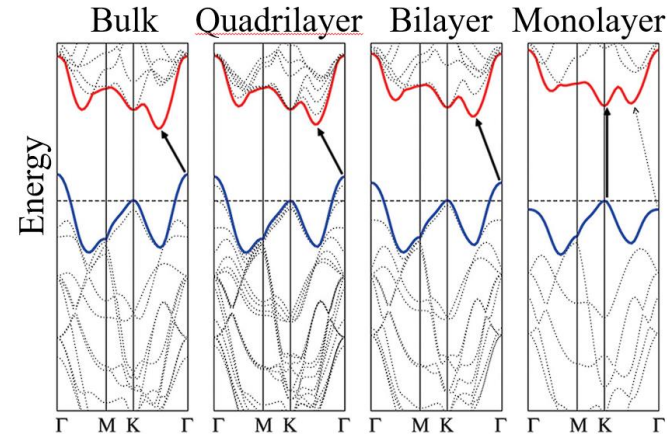
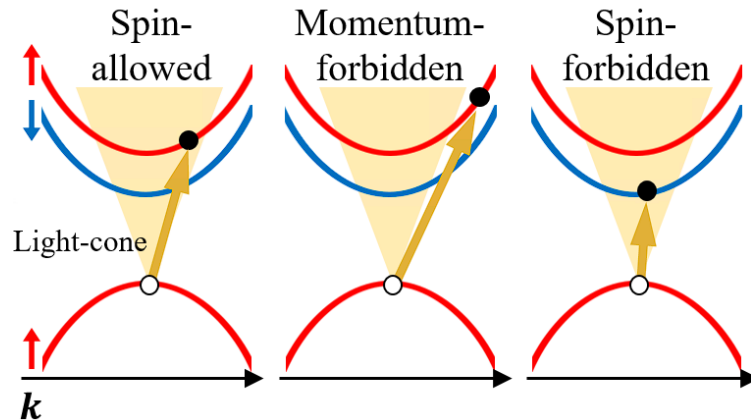
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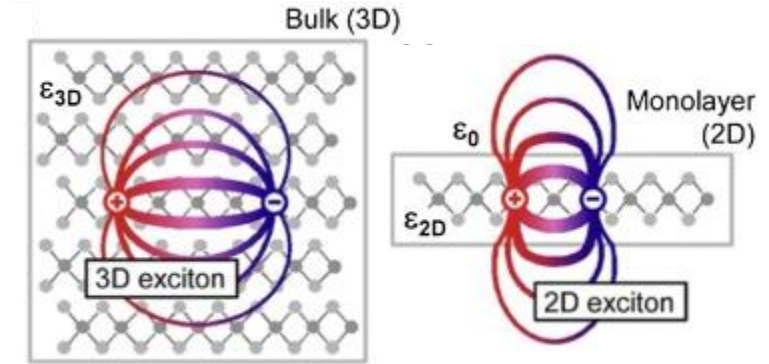
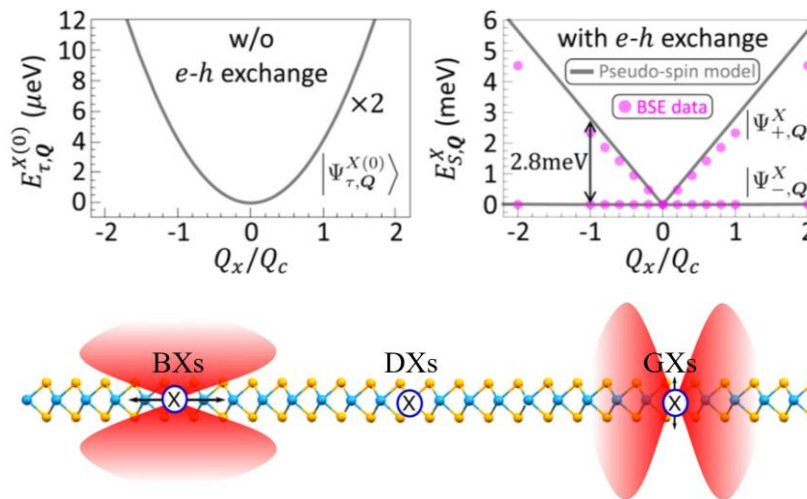
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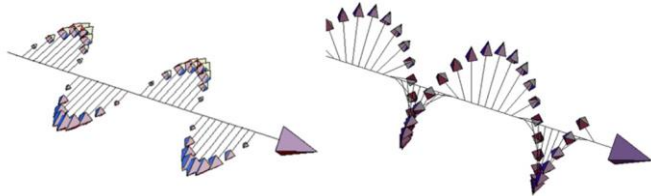
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- Excitons categorized by radiation patterns  
L. M. Schneider *et al.*, *Sci. Rep.* **10**, 8091 (2020)

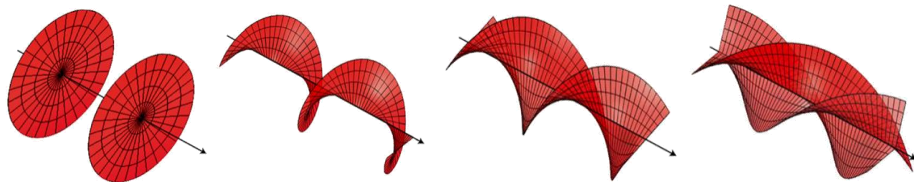
What light information ?

# 1. Introduction

- Angular momentum in light



- Spin angular momentum (SAM)
  - > circular polarizations
  - >  $\sigma\hbar$  per photon ( $\sigma = \pm 1$ )

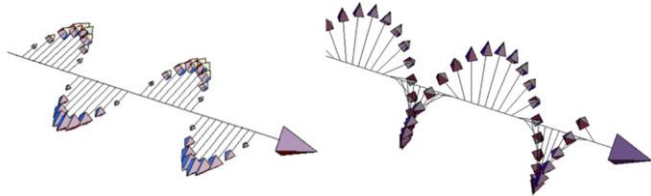


- Orbital angular momentum (OAM)
  - > helical wave front
  - >  $\ell\hbar$  per photon ( $\ell = 0, \pm 1, \pm 2, \pm 3 \dots \dots \pm Z$ )

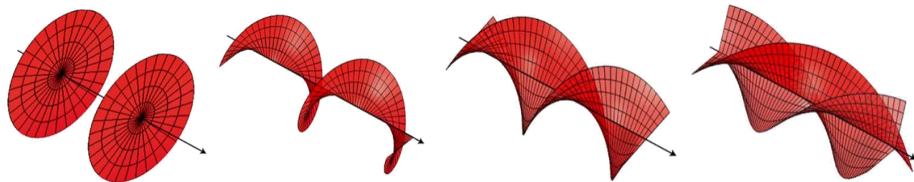
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A. M. Yao & M. J. Padgett, *Adv. Opt. Photonics* **3**, 161 (2011)

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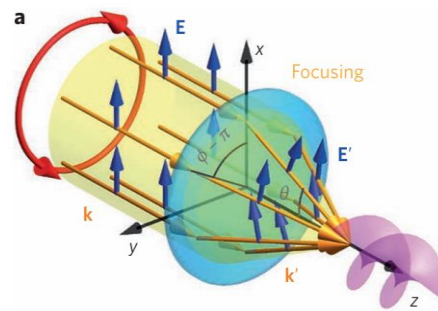
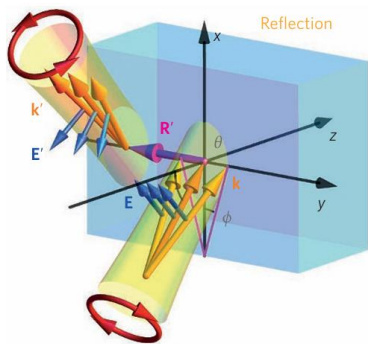


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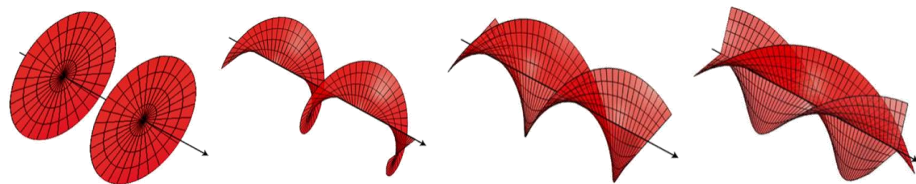
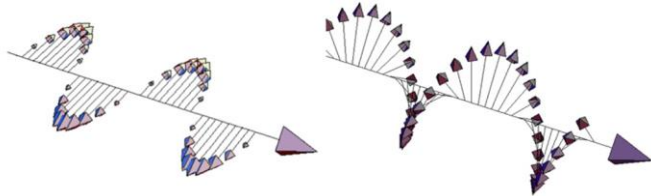


- Spin & Orbital angular momentum interaction (SOI)
  - > circularly polarized laser beam reflected or refracted by medium inhomogeneity
  - > Tight focusing of a paraxial beam
  - > paraxial beam propagates in optical fibres
  - > other .....

- Some cases for SOI  
K. Y. Bliokh *et al.*, *Nat. Photonics* **9**, 796 (2015).

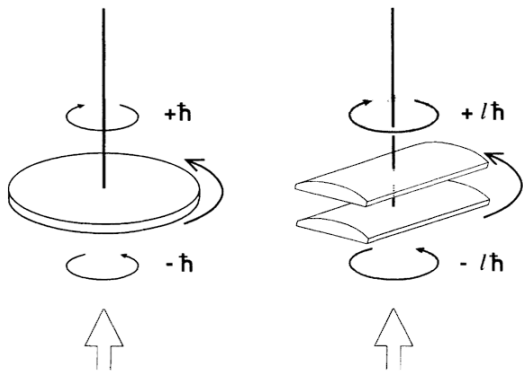
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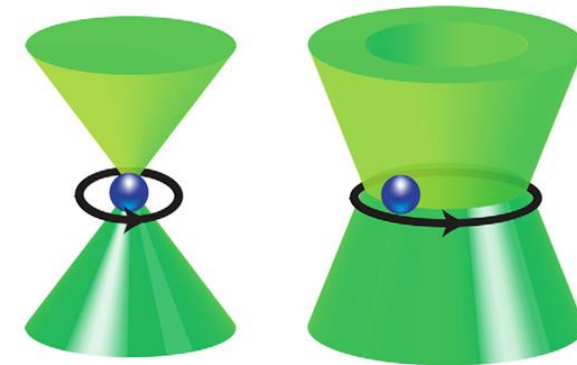
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- Light matter interaction



- Measurements of SAM and OAM  
R. A. Beth, *Phys. Rev.* **50**, 115 (1936)/L. Allen *et al.*, *Phys. Rev. A* **45**, 8185 (1992)

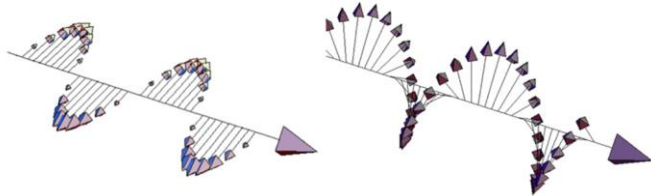
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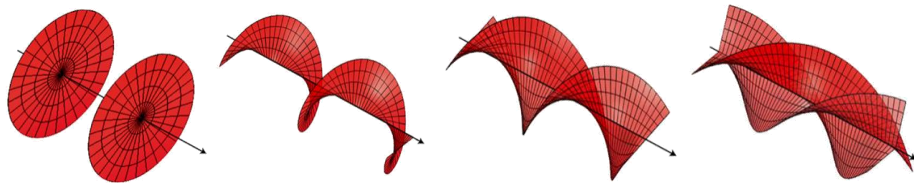
- Transfer of SAM and OAM in optical tweezers  
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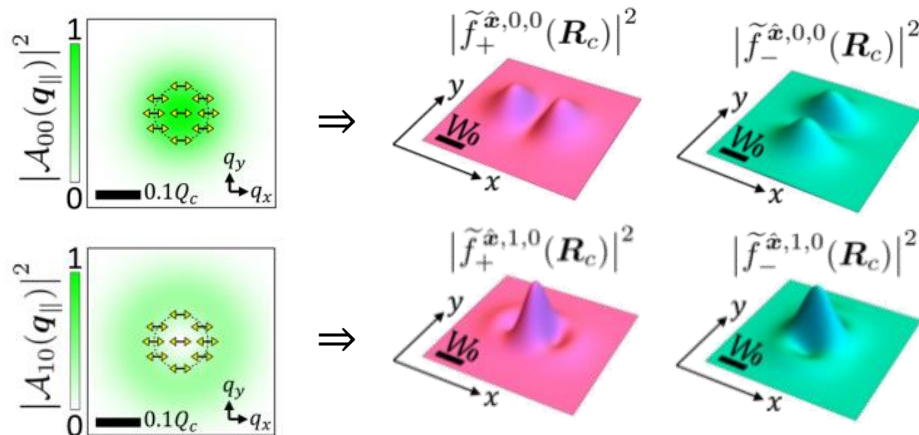


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PHYSICAL REVIEW B **106**, 155304 (2022)

**Tailoring the superposition of finite-momentum valley exciton states in transition-metal dichalcogenide monolayers by using polarized twisted light**

Guan-Hao Peng, Oscar Javier Gomez Sanchez, Wei-Hua Li, Ping-Yuan Lo, and Shun-Jen Cheng  
Department of Electrophysics, National Yang Ming Chiao Tung University, Hsinchu 300, Taiwan

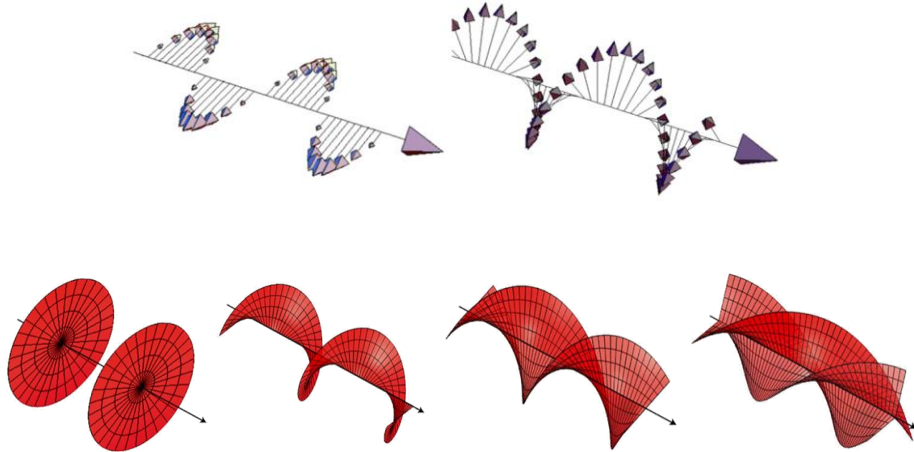
(Received 24 February 2022; revised 23 September 2022; accepted 29 September 2022; published 11 October 2022)

- OAM can influence exciton's wave packet structures



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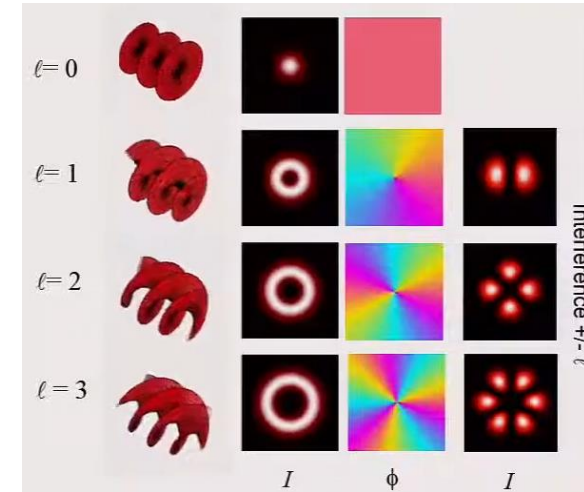
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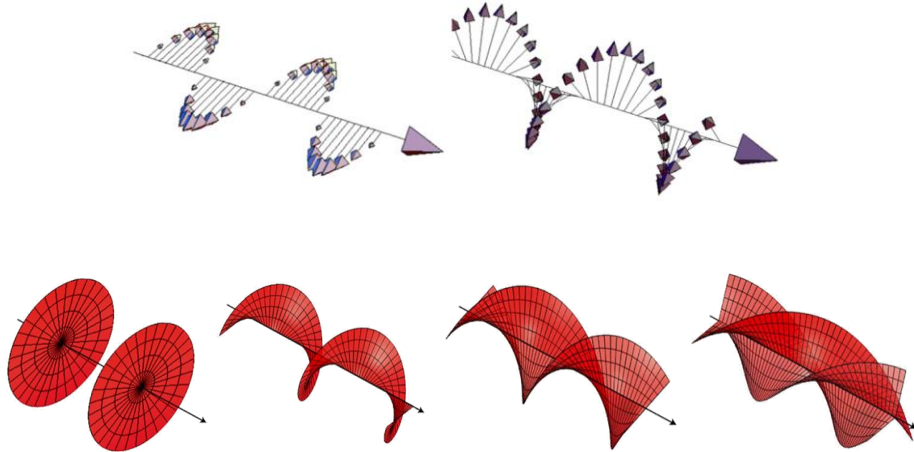


- LG beams and vector vortex beams (transverse plane)
- Reproduced from Miles J. Padgett, International School on Parametric Nonlinear Optics (2015)



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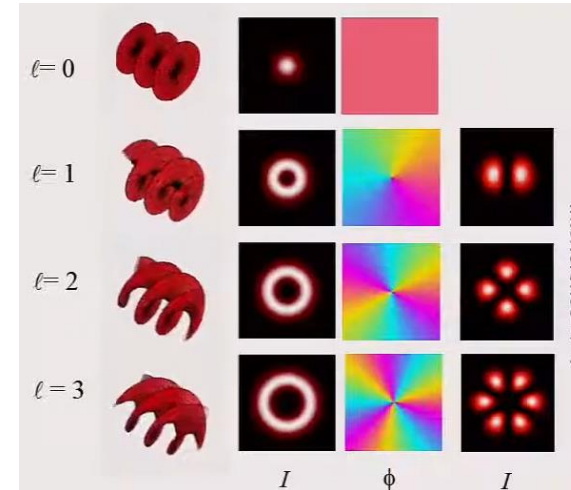
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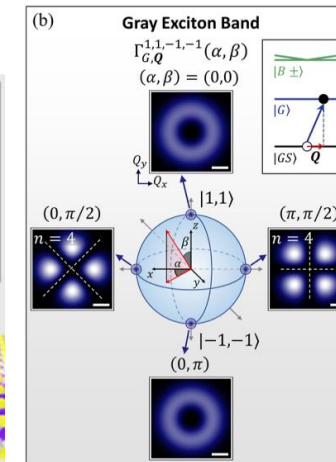
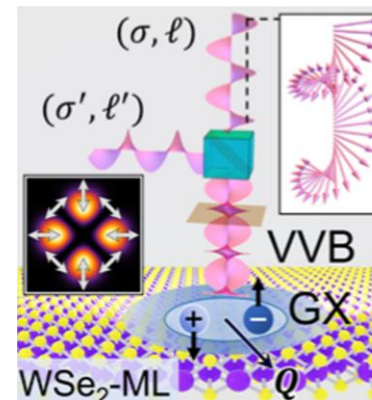
## Enhanced Photo-excitation and Angular-Momentum Imprint of Gray Excitons in WSe<sub>2</sub> Monolayers by Spin–Orbit-Coupled Vector Vortex Beams

Oscar Javier Gomez Sanchez,<sup>§</sup> Guan-Hao Peng,<sup>\*,§</sup> Wei-Hua Li, Ching-Hung Shih, Chao-Hsin Chien, and Shun-Jen Cheng<sup>\*</sup>

- SOI influence optical transition rate through Vector vortex beam



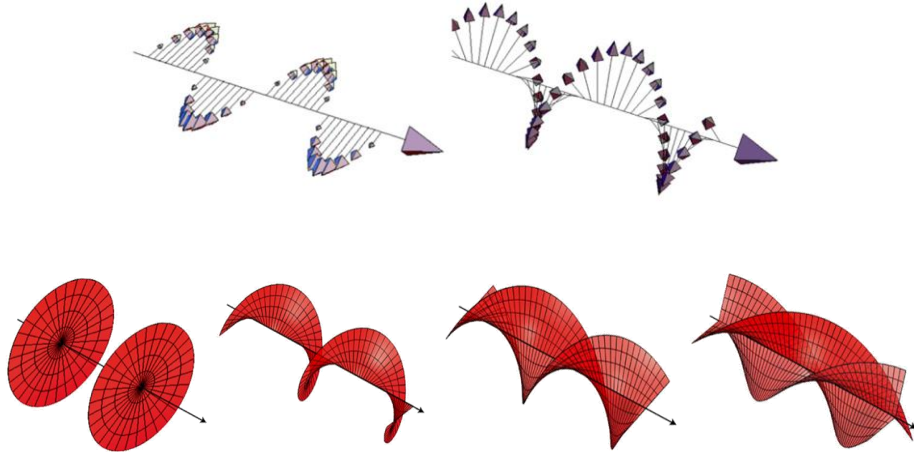
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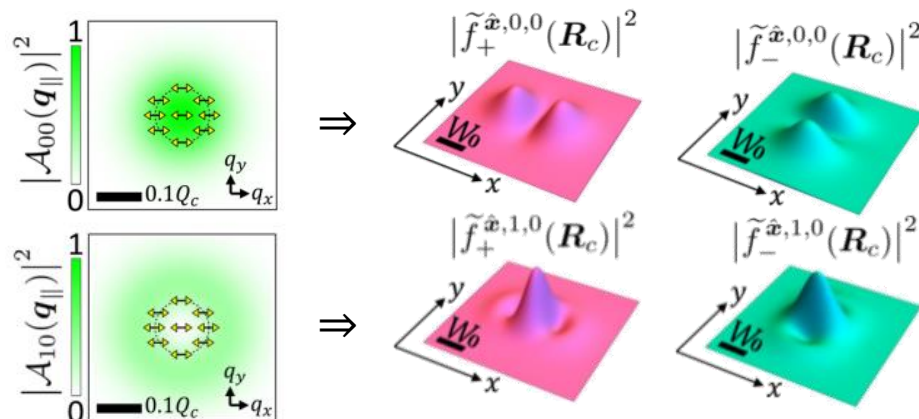
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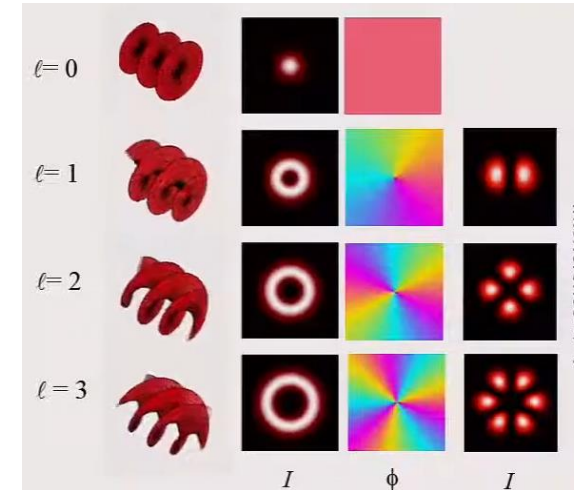


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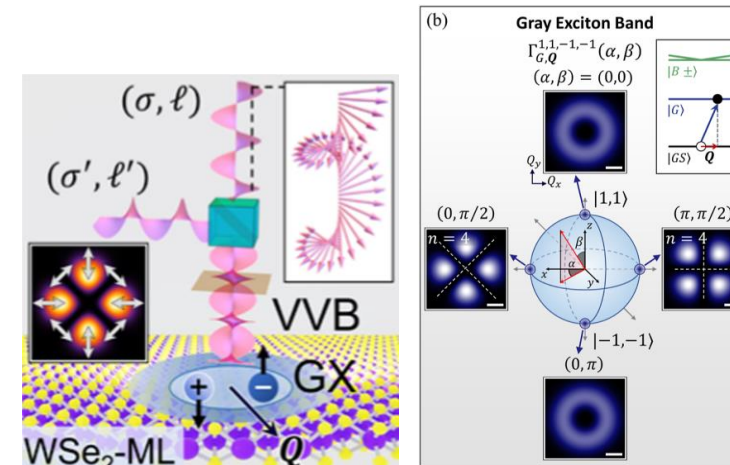
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## 1. Introduction

## 2. Spin-orbital angular momentum interaction (SOI) in light

- Laguerre-Gaussian beams
- Longitudinal component : SOI effect
- Longitudinal component : Vector vortex beams

## 3. Light-matter interaction

## 4. Exciton wave packet

## Summary and outlook

## 2. Laguerre-Gaussian (LG) beam : Derivation

LG beam is solved as vector potential  $\mathbf{A}(\mathbf{r})$  from paraxial Helmholtz equation within **Lorentz gauge**.

**Why?**

Maxwell equations

scalar Helmholtz equation

$$(\nabla^2 + q_0^2)A(\mathbf{r}) = 0$$

**paraxial approximation**

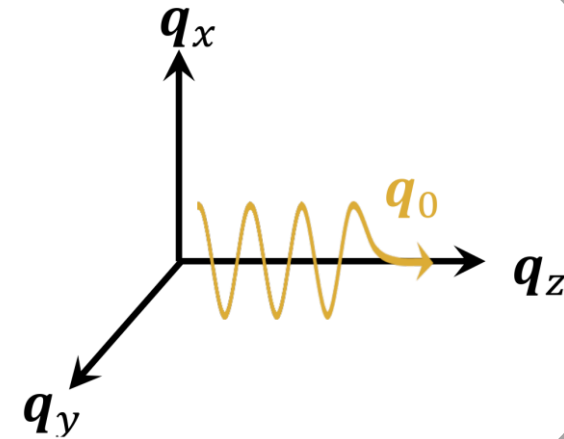
$$\left(\nabla_{\parallel}^2 + \frac{\partial^2}{\partial z^2} + 2iq_0 \frac{\partial}{\partial z}\right)U(\mathbf{r}) = 0$$

Ansatz

Paraxial wave

$$A(\mathbf{r}) = U(\mathbf{r})e^{iq_0 z}$$

wave vector  $\mathbf{q}_z \approx \mathbf{q}_0$



$$\left|\frac{\partial^2}{\partial z^2}U(\mathbf{r})\right| \ll q_0 \left|\frac{\partial}{\partial z}U(\mathbf{r})\right|$$

Approx.

$$\left|\frac{\partial^2}{\partial z^2}U(\mathbf{r})\right| \ll |\nabla_{\parallel}^2 U(\mathbf{r})|$$

paraxial Helmholtz equation

$$\left(\nabla_{\parallel}^2 + 2iq_0 \frac{\partial}{\partial z}\right)U(\mathbf{r}) \approx 0$$

**Fail**

**Coulomb gauge condition**

$$\nabla \cdot \mathbf{A}^C(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} \cdot \nabla_{\parallel} A^C(\mathbf{r})$$

$$= \left(\frac{\partial}{\partial x}\hat{\mathbf{x}} + i\sigma \frac{\partial}{\partial y}\hat{\mathbf{y}}\right)A^C(\mathbf{r}) = 0, \text{ for } \hat{\mathbf{e}}_{\sigma} = \hat{\mathbf{x}} + i\sigma\hat{\mathbf{y}} \\ \sigma = \pm 1$$

eventually constraint the transverse component

$$\nabla_{\parallel}^2 A^C(\mathbf{r}) = \left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)A^C(\mathbf{r}) = 0.$$

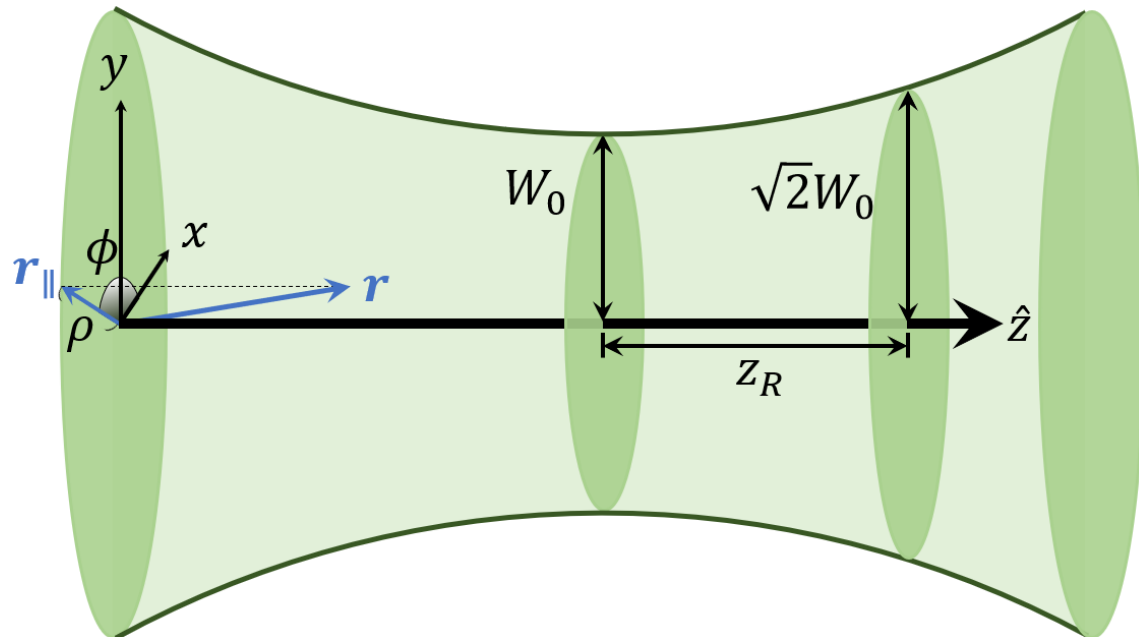
## 2. Laguerre-Gaussian (LG) beam : Distribution

LG beam is solved as vector potential  $\mathbf{A}(\mathbf{r})$  from paraxial Helmholtz equation within Lorentz gauge.

With **long Rayleigh range approximation** :

$$A_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell| + p)!}} L_p^{|\ell|} \left( \frac{2\rho^2}{W_0^2} \right) \left( \frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$



- Topological charge:  
 $\ell \in \mathbb{Z}$   
 (LG beam carries OAM of  $\ell\hbar$ )

- Polarization unit vector :  
 $\hat{\mathbf{e}}_{\sigma} = \hat{\mathbf{x}} + i\sigma\hat{\mathbf{y}}, \quad \sigma = \pm 1$   
 (LG beam carries SAM of  $\sigma\hbar$ )

- Radial mode index: integer  $p \geq 0$
- Beam waist :  $W_0$
- Rayleigh range :  $z_R = \frac{1}{2} q_0 W_0^2$
- Constant amplitude :  $A_0^{\text{LG}}$
- Generalized Laguerre polynomial :  

$$L_p^{|\ell|}(x) = \sum_{m=0}^p (-1)^m \frac{(|\ell|+p)!}{(p-m)! (|\ell|+m)! m!} x^m$$



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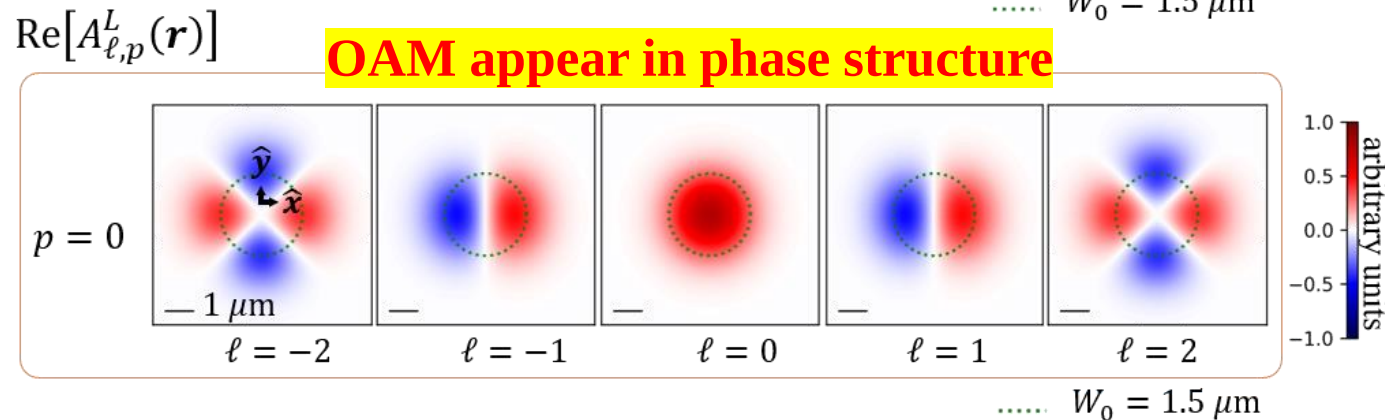
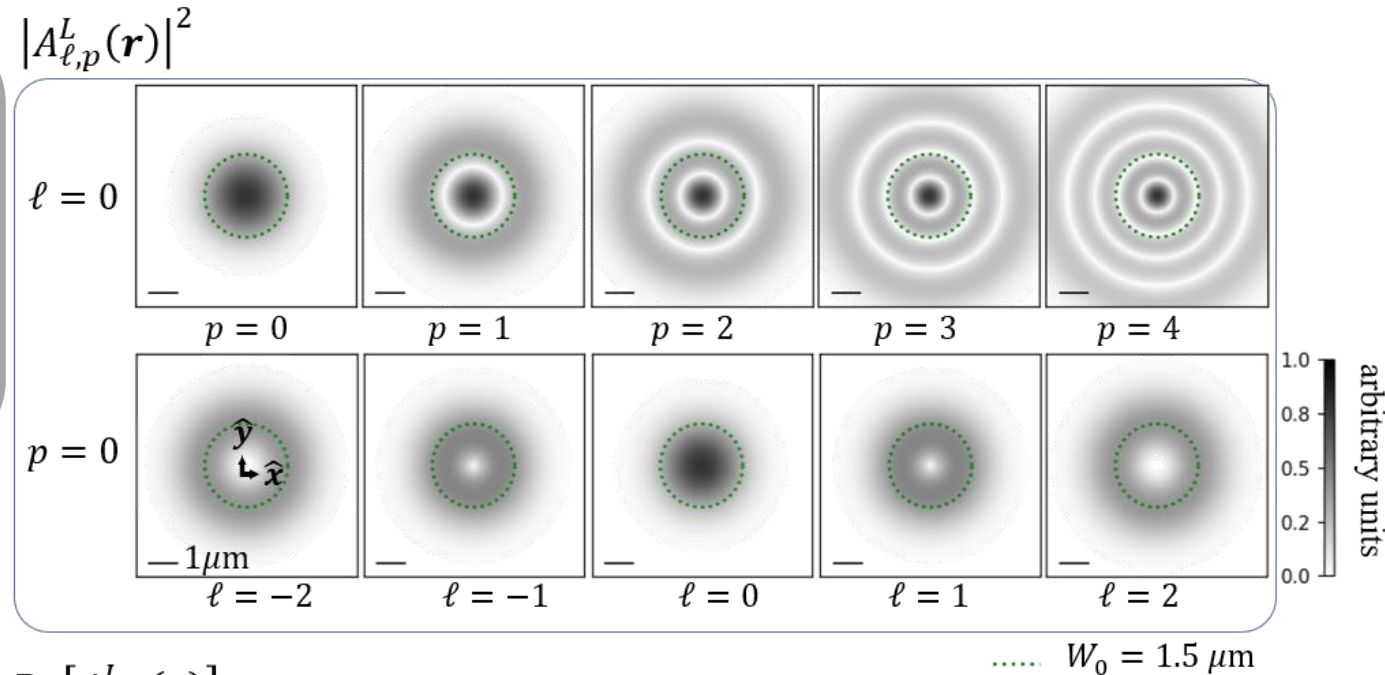
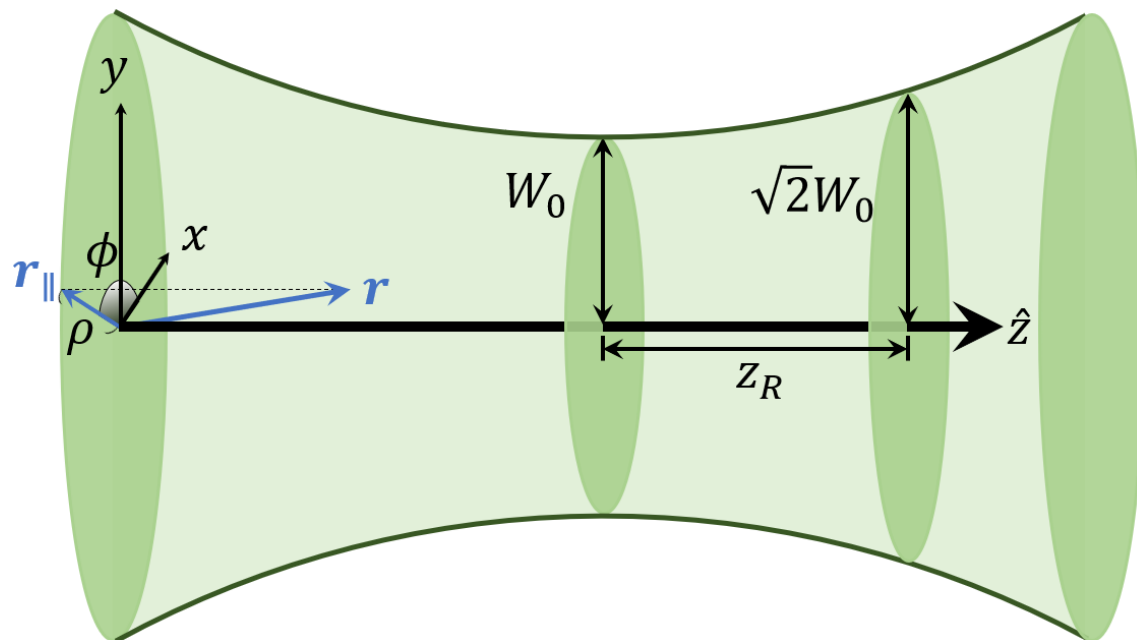
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**Separated SAM & OAM**

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell| + p)!}} L_p^{|\ell|} \left( \frac{2\rho^2}{W_0^2} \right) \left( \frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

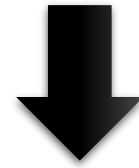




## 2. Laguerre-Gaussian (LG) beam : Requirement

LG beam in real space / Lorentz gauge :

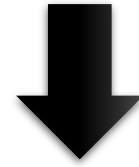
$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r})$$



LG beam in angular spectrum / Coulomb gauge :

$$\mathcal{A}_{\sigma,\ell,p}^C(q_{\parallel})$$

**What we really need**



Theory of light matter interaction

$$\Gamma^{S,Q} = \frac{2\pi}{\hbar} |\tilde{M}^{S,Q}|^2 \rho^L(\hbar\omega = E^{S,Q})$$

$$\tilde{M}^{S,Q} \approx \frac{E_g}{2i\hbar} \delta_{q_{\parallel}Q} \mathcal{A}^C(q_{\parallel}) \cdot \mathbf{D}^{S,Q*}$$

## 2. Laguerre-Gaussian (LG) beam : Angular spectrum

LG beam in Lorentz gauge / **real space** :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell|+p)!}} L_p^{|\ell|} \left( \frac{2\rho^2}{W_0^2} \right) \left( \frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

LG beam in Lorentz gauge / **angular spectrum** :

$$\mathcal{A}_{\sigma,\ell,p}^L(\mathbf{q}_{\parallel}) = \hat{\mathbf{e}}_{\sigma} \mathcal{A}_{\ell,p}^L(\mathbf{q}_{\parallel}) \approx \hat{\mathbf{e}}_{\sigma} \tilde{F}_{|\ell|,p}(q_{\parallel}) e^{i\ell\phi_{q_{\parallel}}}$$

$$\tilde{F}_{|\ell|,p}(q_{\parallel}) = \frac{2\pi(-i)^{|\ell|}}{\Omega} \mathcal{H}_{|\ell|}[f_{\ell,p}(\rho)]$$

Hankel transform of order  $|\ell|$

**Method**

Goal : expanded **in plane-wave basis**

$$\mathbf{A}(\mathbf{r}) = \sum_{\mathbf{q}} \mathcal{A}(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{r}}$$

Define Fourier transformation

$$f_{\ell,p}(\rho) e^{i\ell\phi} = u_{\ell,p}(\mathbf{r}_{\parallel}) = \frac{1}{(2\pi)^2} \int d^2 \mathbf{q}_{\parallel} F_{\ell,p}^{\text{LG}}(q_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}}$$

$$F_{\ell,p}^{\text{LG}}(q_{\parallel}) = \int d^2 \mathbf{r}_{\parallel} u_{\ell,p}(\mathbf{r}_{\parallel}) e^{-i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}}$$

Results

$$\mathcal{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \sum_{\mathbf{q}_{\parallel}} \mathcal{A}_{\sigma,\ell,p}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z}$$

## 2. Laguerre-Gaussian (LG) beam : Angular spectrum

LG beam in Lorentz gauge / **real space** :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell|+p)!}} L_p^{|\ell|} \left( \frac{2\rho^2}{W_0^2} \right) \left( \frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

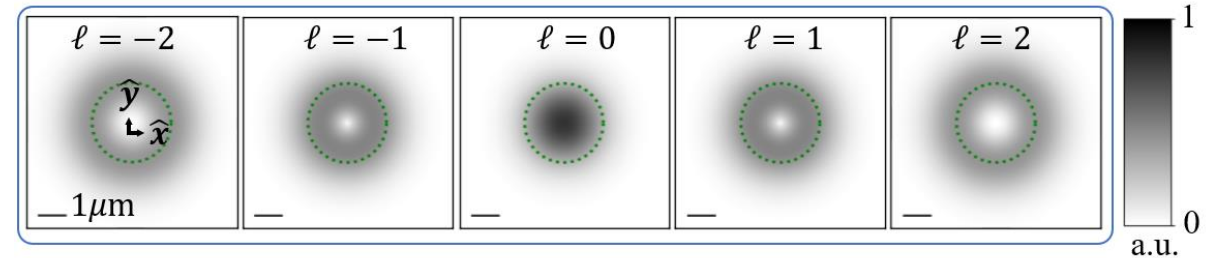
LG beam in Lorentz gauge / **angular spectrum** :

$$\mathcal{A}_{\sigma,\ell,p}^L(q_{\parallel}) = \hat{\mathbf{e}}_{\sigma} \mathcal{A}_{\ell,p}^L(q_{\parallel}) \approx \hat{\mathbf{e}}_{\sigma} \tilde{F}_{|\ell|,p}(q_{\parallel}) e^{i\ell\phi_{q_{\parallel}}}$$

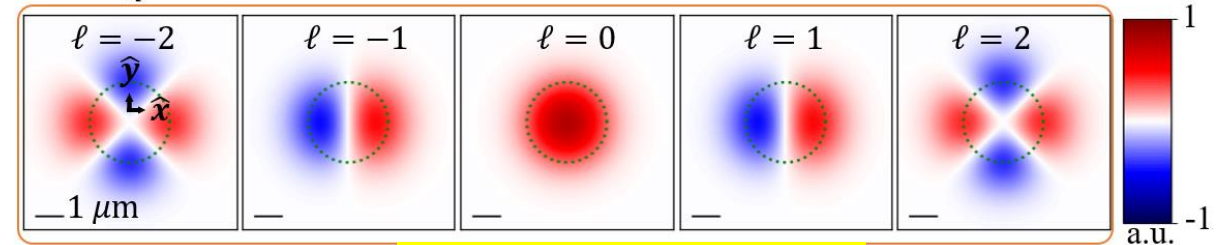
$$\tilde{F}_{|\ell|,p}(q_{\parallel}) = \frac{2\pi(-i)^{|\ell|}}{\Omega} \mathcal{H}_{|\ell|}[f_{\ell,p}(\rho)]$$

Hankel transform of order  $|\ell|$

$$|A_{\ell,p=0}^L(\mathbf{r})|^2$$

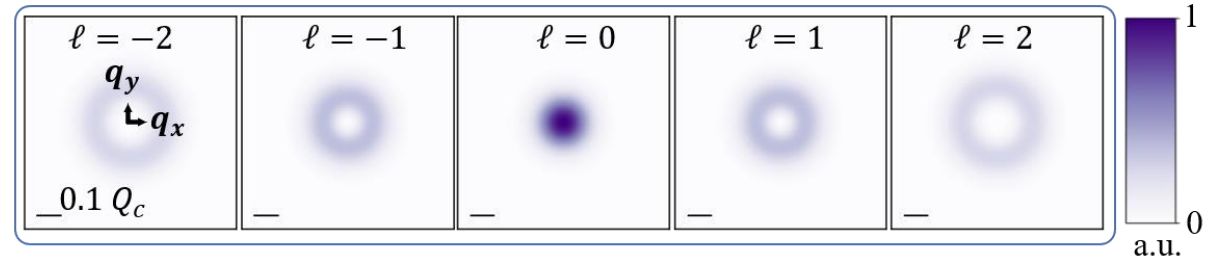


$$\text{Re}[A_{\ell,p=0}^L(\mathbf{r})]$$

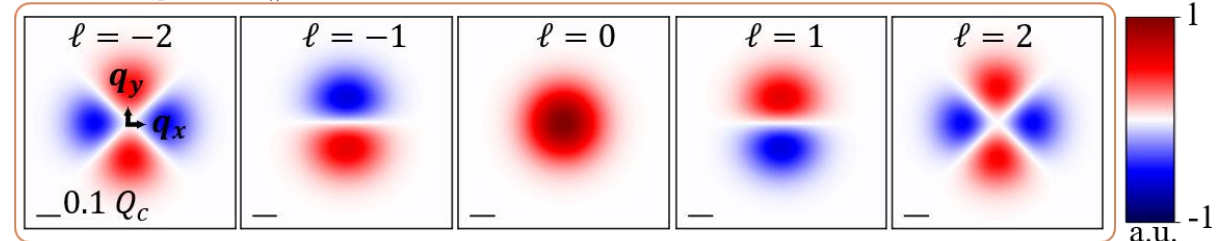


**Similar structure**

$$|\mathcal{A}_{\ell,p=0}^L(q_{\parallel})|^2$$



$$\text{Re}[\mathcal{A}_{\ell,p=0}^L(q_{\parallel})]$$



## 2. Laguerre-Gaussian (LG) beam : Coulomb gauge

LG beam in **Lorentz gauge** / real space :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell|+p)!}} L_p^{|\ell|} \left( \frac{2\rho^2}{W_0^2} \right) \left( \frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

LG beam in **Coulomb gauge** / real space:

$$\mathbf{A}_{\sigma,\ell,p}(r) \equiv \mathbf{A}_{\sigma,\ell,p}^C(r) \approx \tilde{A}_{\ell,p}^{\parallel}(r) \hat{\mathbf{e}}_{\sigma} + \tilde{A}_{\sigma,\ell,p}^z(r) \hat{\mathbf{z}}$$

Transverse component :

$$\tilde{A}_{\ell,p}^{\parallel}(r) \approx f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

Longitudinal component :

$$\tilde{A}_{\sigma,\ell,p}^z(r) \approx \frac{i}{\sqrt{2}q_0\rho} \left[ (|\ell| + \sigma\ell) - \frac{2\rho^2}{W_0^2} \right] f_{\ell,p}(\rho) e^{i(\sigma+\ell)\phi} e^{iq_0 z}$$

**Additional component**

**Method**

Relation from Maxwell's equations

$$\mathbf{A}_{\sigma,\ell,p}^C(r) = \mathbf{A}_{\sigma,\ell,p}^L(r) + \frac{\nabla \left( \nabla \cdot \mathbf{A}_{\sigma,\ell,p}^L(r) \right)}{q_0^2}$$

## 2. Laguerre-Gaussian (LG) beam : SOI appear

LG beam in **Lorentz gauge** / real space :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell|+p)!}} L_p^{|\ell|} \left( \frac{2\rho^2}{W_0^2} \right) \left( \frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

—  $|\tilde{A}_{\ell,p=0}^{\parallel}(r)|$   
 —  $|\tilde{A}_{\sigma=1,\ell,p=0}^z(r)|$   
 —  $|\tilde{A}_{\sigma=-1,\ell,p=0}^z(r)|$

|                   | Transverse     | Longitudinal             |
|-------------------|----------------|--------------------------|
| Between $\pm\ell$ | symmetric      | Asymmetric<br><b>SOI</b> |
| Focused magnitude | merely changed | Larger                   |

LG beam in **Coulomb gauge** / real space:

$$\mathbf{A}_{\sigma,\ell,p}(r) \equiv \mathbf{A}_{\sigma,\ell,p}^C(r) \approx \tilde{A}_{\ell,p}^{\parallel}(r) \hat{\mathbf{e}}_{\sigma} + \tilde{A}_{\sigma,\ell,p}^z(r) \hat{\mathbf{z}}$$

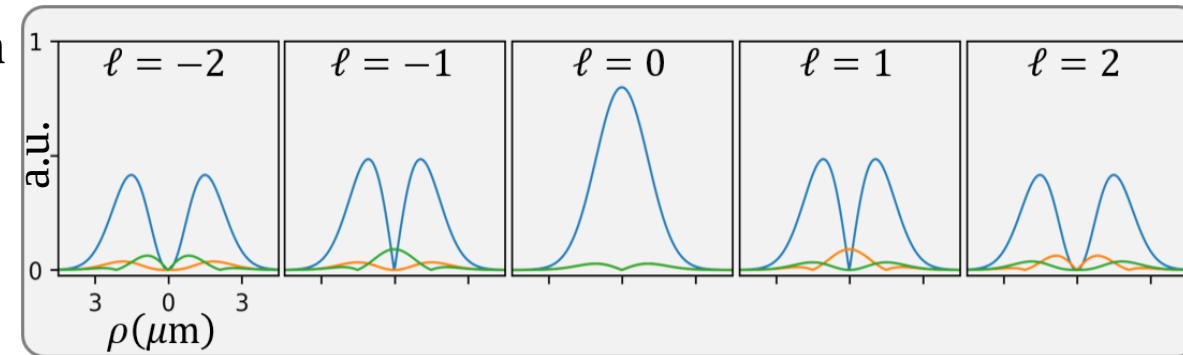
Transverse component :

$$\tilde{A}_{\ell,p}^{\parallel}(r) \approx f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

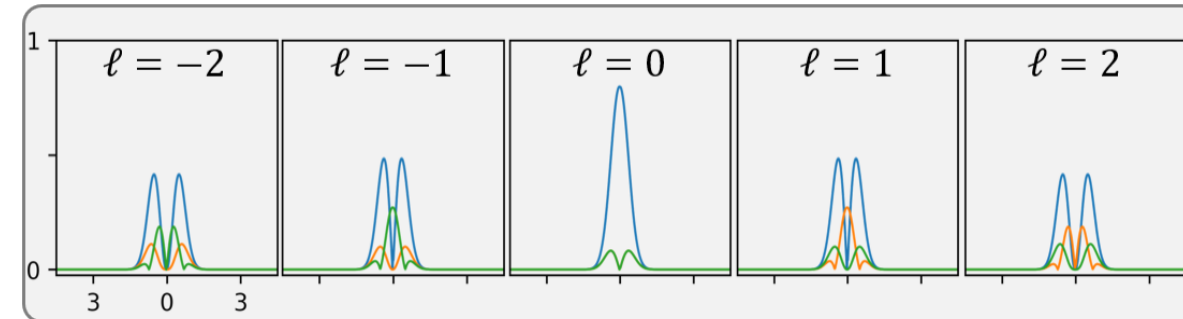
Longitudinal component :

$$\tilde{A}_{\sigma,\ell,p}^z(r) \approx \frac{i}{\sqrt{2}q_0\rho} \left[ (|\ell| + \text{SOI}) - \frac{2\rho^2}{W_0^2} \right] f_{\ell,p}(\rho) e^{i(\sigma+\ell)\phi} e^{iq_0 z}$$

$W_0 = 1.5\mu\text{m}$



$W_0 = 0.5\mu\text{m}$



## 2. Longitudinal component : SOI effect

LG beam solutions:

Real / Lorentz :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

Angular / Lorentz :

$$\mathcal{A}_{\sigma,\ell,p}^L(\mathbf{q}_{\parallel}) = \hat{\mathbf{e}}_{\sigma} \mathcal{A}_{\ell,p}^L(\mathbf{q}_{\parallel}) \approx \hat{\mathbf{e}}_{\sigma} \tilde{F}_{|\ell|,p}(\mathbf{q}_{\parallel}) e^{i\ell\phi_{\mathbf{q}_{\parallel}}}$$

Real / Coulomb :

$$\mathbf{A}_{\sigma,\ell,p}(\mathbf{r}) \equiv \mathbf{A}_{\sigma,\ell,p}^C(\mathbf{r}) \approx \tilde{A}_{\ell,p}^{\parallel}(\mathbf{r}) \hat{\mathbf{e}}_{\sigma} + \tilde{A}_{\sigma,\ell,p}^z(\mathbf{r}) \hat{\mathbf{z}}$$

$$\mathcal{A}_{\sigma,\ell,p}^C(\mathbf{q}_{\parallel}) \quad \text{What we really need}$$

LG beam in **angular spectrum** / **Coulomb gauge** :

$$\mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) \equiv \mathcal{A}_{\sigma,\ell,p=0}^C(\mathbf{q}_{\parallel}) \approx \tilde{\mathcal{A}}_{\ell,p}^{\parallel}(\mathbf{q}_{\parallel}) \hat{\mathbf{e}}_{\sigma} + \tilde{\mathcal{A}}_{\sigma,\ell,p}^z(\mathbf{q}_{\parallel}) \hat{\mathbf{z}}$$

Transverse component :

$$\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \equiv \tilde{\mathcal{A}}_{\ell,p=0}^{\parallel}(\mathbf{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(\mathbf{q}_{\parallel}) e^{i\ell\phi_{\mathbf{q}_{\parallel}}}$$

Longitudinal component :

$$\tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel}) \equiv \tilde{\mathcal{A}}_{\sigma,\ell,p=0}^z(\mathbf{q}_{\parallel}) \approx -\frac{\sin \theta_{\mathbf{q}}}{\sqrt{2}} \tilde{F}_{|\ell|}(\mathbf{q}_{\parallel}) e^{i(\sigma+\ell)\phi_{\mathbf{q}_{\parallel}}} \quad \text{SOI}$$

Method

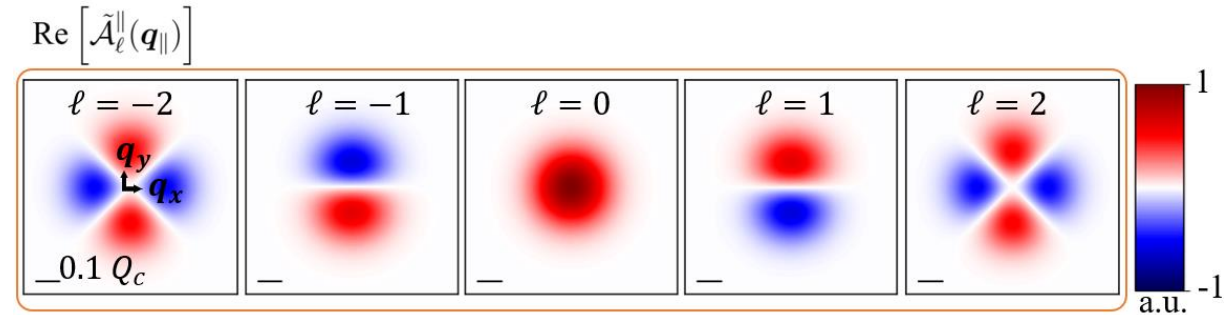
expanded in plane-wave basis

$$\mathbf{A}_{\sigma,\ell,p}^C(\mathbf{r}) = \frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \mathcal{A}_{\sigma,\ell,p}^C(\mathbf{q}_{\parallel}) e^{i\mathbf{q} \cdot \mathbf{r}}$$

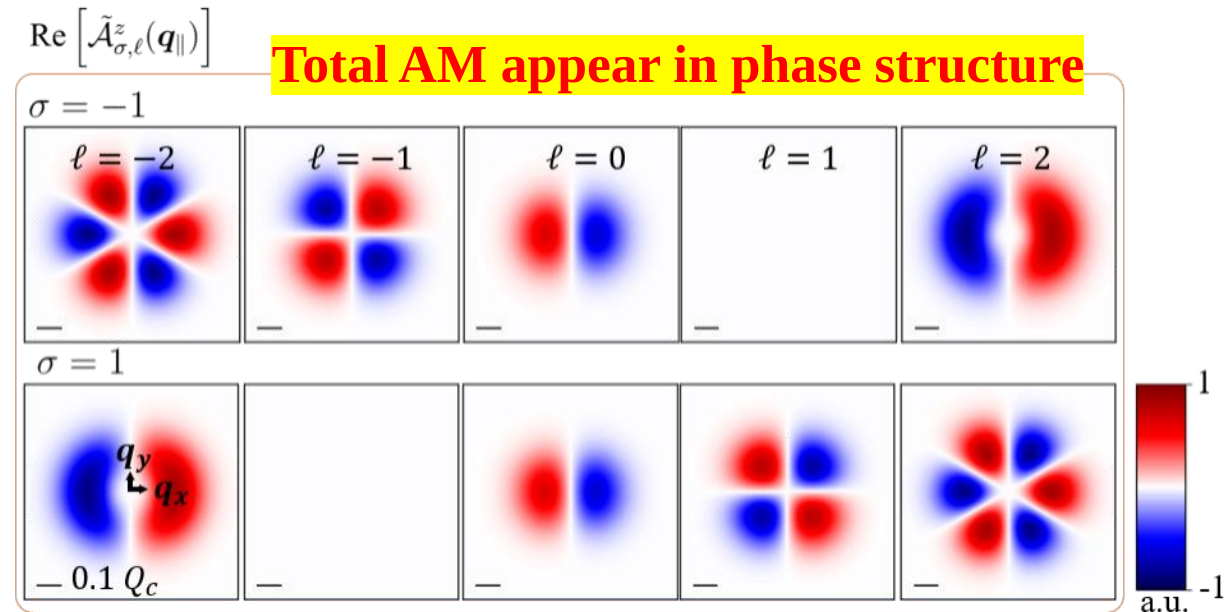


## 2. Longitudinal component : SOI effect

Transverse component :



Longitudinal component :



$$\mathcal{A}_{\sigma,\ell,p}^c(\mathbf{q}_{\parallel})$$

LG beam in **angular spectrum** / **Coulomb gauge** :

$$\mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) \equiv \mathcal{A}_{\sigma,\ell,p=0}^c(\mathbf{q}_{\parallel}) \approx \tilde{\mathcal{A}}_{\ell,p}^{\parallel}(\mathbf{q}_{\parallel}) \hat{\mathbf{e}}_{\sigma} + \tilde{\mathcal{A}}_{\sigma,\ell,p}^z(\mathbf{q}_{\parallel}) \hat{\mathbf{z}}$$

Transverse component :

$$\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \equiv \tilde{\mathcal{A}}_{\ell,p=0}^{\parallel}(\mathbf{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\ell\phi_{\mathbf{q}_{\parallel}}}$$

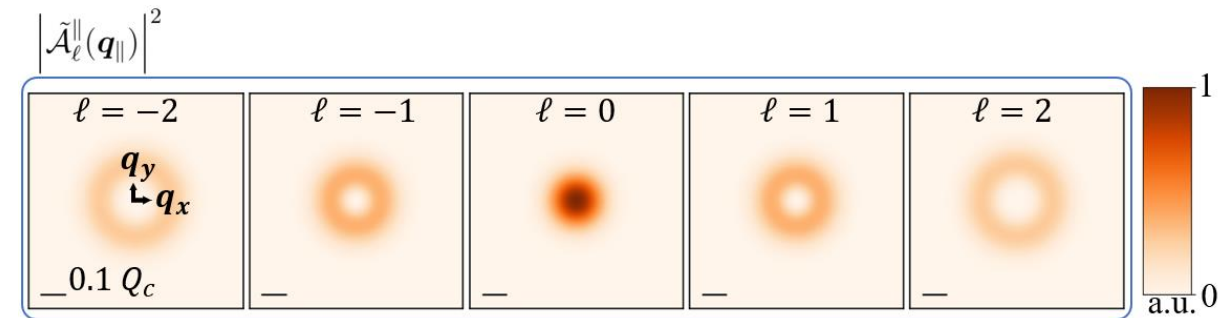
Longitudinal component :

$$\tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel}) \equiv \tilde{\mathcal{A}}_{\sigma,\ell,p=0}^z(\mathbf{q}_{\parallel}) \approx -\frac{\sin \theta_{\mathbf{q}}}{\sqrt{2}} \tilde{F}_{|\ell|}(q_{\parallel}) e^{i(\sigma+\ell)\phi_{\mathbf{q}_{\parallel}}}$$

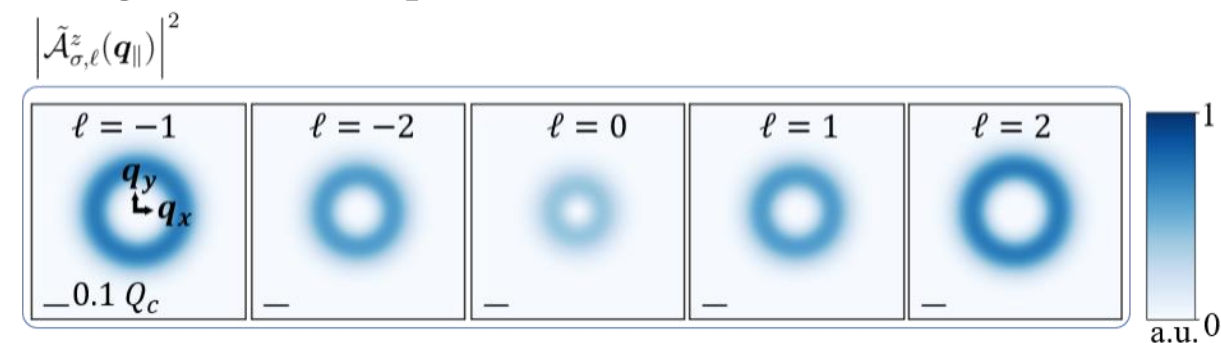
**SOI**

## 2. Longitudinal component : SOI effect

Transverse component :



Longitudinal component :



**Same radial structure for  $\sigma = \pm 1$**

$$\mathcal{A}_{\sigma,\ell,p}^c(\mathbf{q}_{\parallel})$$

LG beam in **angular spectrum** / **Coulomb gauge** :

$$\mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) \equiv \mathcal{A}_{\sigma,\ell,p=0}^c(\mathbf{q}_{\parallel}) \approx \tilde{\mathcal{A}}_{\ell,p}^{\parallel}(\mathbf{q}_{\parallel}) \hat{\mathbf{e}}_{\sigma} + \tilde{\mathcal{A}}_{\sigma,\ell,p}^z(\mathbf{q}_{\parallel}) \hat{\mathbf{z}}$$

Transverse component :

$$\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \equiv \tilde{\mathcal{A}}_{\ell,p=0}^{\parallel}(\mathbf{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\ell\phi_{\mathbf{q}_{\parallel}}}$$

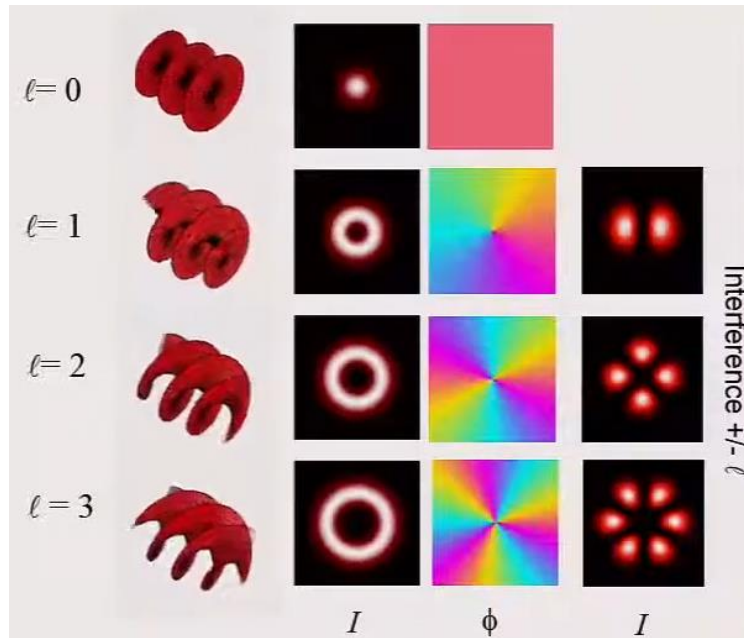
Longitudinal component :

$$\tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel}) \equiv \tilde{\mathcal{A}}_{\sigma,\ell,p=0}^z(\mathbf{q}_{\parallel}) \approx -\frac{\sin \theta_{\mathbf{q}}}{\sqrt{2}} \tilde{F}_{|\ell|}(q_{\parallel}) e^{i(\sigma+\ell)\phi_{\mathbf{q}_{\parallel}}}$$

**SOI**

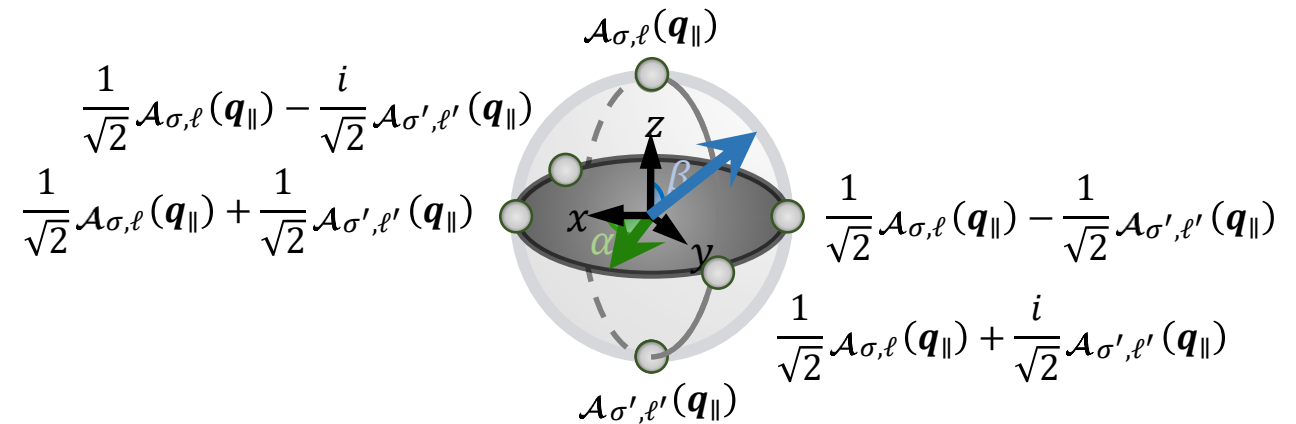
## 2. Vector vortex beams (VVB)

Forming vector vortex beams (VVB) by Superimposing LG beams



- LG beams and vector vortex beams (transverse plane)  
Reproduced from Miles J. Padgett, International School on  
Parametric Nonlinear Optics (2015)

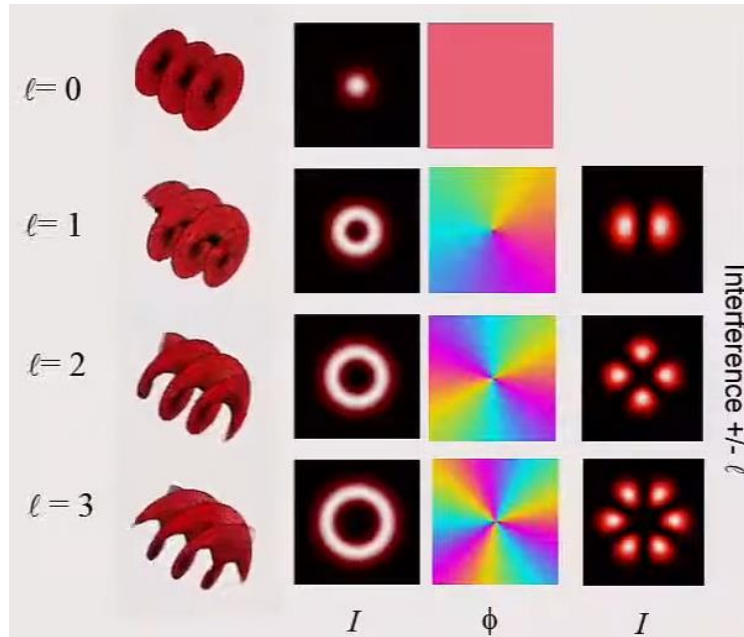
$$\mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) = \cos\left(\frac{\beta}{2}\right) \mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) + e^{i\alpha} \sin\left(\frac{\beta}{2}\right) \mathcal{A}_{\sigma',\ell'}(\mathbf{q}_{\parallel})$$



- Poincaré sphere for OAM states

## 2. Vector vortex beams (VVB)

Forming vector vortex beams (VVB) by Superimposing LG beams



- LG beams and vector vortex beams (transverse plane)  
Reproduced from Miles J. Padgett, International School on  
Parametric Nonlinear Optics (2015)

$$\mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) = \cos\left(\frac{\beta}{2}\right) \mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) + e^{i\alpha} \sin\left(\frac{\beta}{2}\right) \mathcal{A}_{\sigma',\ell'}(\mathbf{q}_{\parallel})$$

$$= \mathcal{A}_{\sigma,\ell,\sigma',\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta) + \mathcal{A}_{\sigma,\ell,\sigma',\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta)$$

$$\text{Define } \Delta\ell \equiv \ell' - \ell \quad 2\bar{\ell} \equiv \ell' + \ell \quad \Delta\sigma \equiv \sigma' - \sigma \quad 2\bar{\sigma} \equiv \sigma' + \sigma$$

$$= \mathcal{A}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta) + \mathcal{A}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}^z(\mathbf{q}_{\parallel}; \alpha, \beta)$$

## 2. Transverse component : Vector vortex beams (VVB)

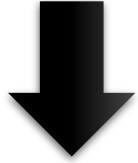
Transverse component of VVB:

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta\sigma, \bar{\sigma}}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta) = \mathcal{A}_{\Delta\ell, \bar{\ell}}^x(\mathbf{q}_{\parallel}; \alpha, \beta)\hat{x} + \mathcal{A}_{\Delta\ell, \bar{\ell}}^y(\mathbf{q}_{\parallel}; \alpha, \beta)\hat{y}$$

## 2. Transverse component : Vector vortex beams (VVB)

Transverse component of VVB:

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta\sigma, \bar{\sigma}}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta) = \mathcal{A}_{\Delta\ell, \bar{\ell}}^x(\mathbf{q}_{\parallel}; \alpha, \beta)\hat{\mathbf{x}} + \mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta\sigma, \bar{\sigma}}^y(\mathbf{q}_{\parallel}; \alpha, \beta)\hat{\mathbf{y}}$$



Transverse component of VVB:  $\Delta\sigma = -2, \bar{\sigma} = 0 / \beta = \frac{\pi}{2}$

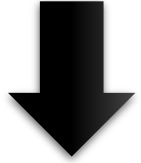
$$\begin{aligned} \mathcal{A}_{\Delta\ell, \bar{\ell}, -2, 0}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) &= \frac{1}{2} e^{i(\bar{\ell}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \cos\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}} + \alpha)\right] \left[ \tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) + \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel}) \right] - i \sin\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}} + \alpha)\right] \left[ \tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) - \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel}) \right] \right\} \hat{\mathbf{x}} \\ &\quad + \frac{1}{2} e^{i(\bar{\ell}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \sin\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}} + \alpha)\right] \left[ \tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) + \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel}) \right] + i \sin\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}} + \alpha + \pi)\right] \left[ \tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) - \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel}) \right] \right\} \hat{\mathbf{y}} \end{aligned}$$



## 2. Transverse component : Vector vortex beams (VVB)

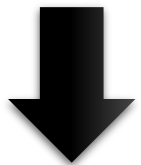
Transverse component of VVB:

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta\sigma, \bar{\sigma}}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta) = \mathcal{A}_{\Delta\ell, \bar{\ell}}^x(\mathbf{q}_{\parallel}; \alpha, \beta)\hat{\mathbf{x}} + \mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta\sigma, \bar{\sigma}}^y(\mathbf{q}_{\parallel}; \alpha, \beta)\hat{\mathbf{y}}$$



Transverse component of VVB:  $\Delta\sigma = -2, \bar{\sigma} = 0 / \beta = \frac{\pi}{2}$

$$\begin{aligned} \mathcal{A}_{\Delta\ell, \bar{\ell}, -2, 0}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) &= \frac{1}{2} e^{i(\bar{\ell}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \cos\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}} + \alpha)\right] [\tilde{F}_{|\bar{\ell}-\frac{\Delta\ell}{2}|}(q_{\parallel}) + \tilde{F}_{|\bar{\ell}+\frac{\Delta\ell}{2}|}(q_{\parallel})] - i \sin\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}} + \alpha)\right] [\tilde{F}_{|\bar{\ell}-\frac{\Delta\ell}{2}|}(q_{\parallel}) - \tilde{F}_{|\bar{\ell}+\frac{\Delta\ell}{2}|}(q_{\parallel})] \right\} \hat{\mathbf{x}} \\ &\quad + \frac{1}{2} e^{i(\bar{\ell}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \sin\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}} + \alpha)\right] [\tilde{F}_{|\bar{\ell}-\frac{\Delta\ell}{2}|}(q_{\parallel}) + \tilde{F}_{|\bar{\ell}+\frac{\Delta\ell}{2}|}(q_{\parallel})] + i \sin\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}} + \alpha + \pi)\right] [\tilde{F}_{|\bar{\ell}-\frac{\Delta\ell}{2}|}(q_{\parallel}) - \tilde{F}_{|\bar{\ell}+\frac{\Delta\ell}{2}|}(q_{\parallel})] \right\} \hat{\mathbf{y}} \end{aligned}$$



$$\text{If } \tilde{F}_{|\bar{\ell}-\frac{\Delta\ell}{2}|}(q_{\parallel}) = \tilde{F}_{|\bar{\ell}+\frac{\Delta\ell}{2}|}(q_{\parallel})$$

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, -2, 0}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = \tilde{F}_{|\bar{\ell} \pm \frac{\Delta\ell}{2}|}(q_{\parallel}) e^{i(\bar{\ell}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \cos\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}} + \alpha)\right] \hat{\mathbf{x}} + \sin\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}} + \alpha)\right] \hat{\mathbf{y}} \right\}$$

**Doughnut shape**

**Linear polarization**

## 2. Transverse component : Vector vortex beams (VVB)

Transverse component of VVB:

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, -2, 0}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = \tilde{F}_{|\bar{\ell} \pm \frac{\Delta\ell}{2}|}(q_{\parallel}) e^{i(\bar{\ell}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \cos\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}} + \alpha)\right] \hat{\mathbf{x}} + \sin\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}} + \alpha)\right] \hat{\mathbf{y}} \right\}$$

**Doughnut shape**

**Linear polarization**

Conditions for  $\tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) = \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel})$  in LG beams  $\Rightarrow \left| \bar{\ell} - \frac{\Delta\ell}{2} \right| = |\ell| = |\ell'| = \left| \bar{\ell} + \frac{\Delta\ell}{2} \right|$

$\Rightarrow$

1.  $\ell = \ell'$
2.  $\ell = -\ell'$

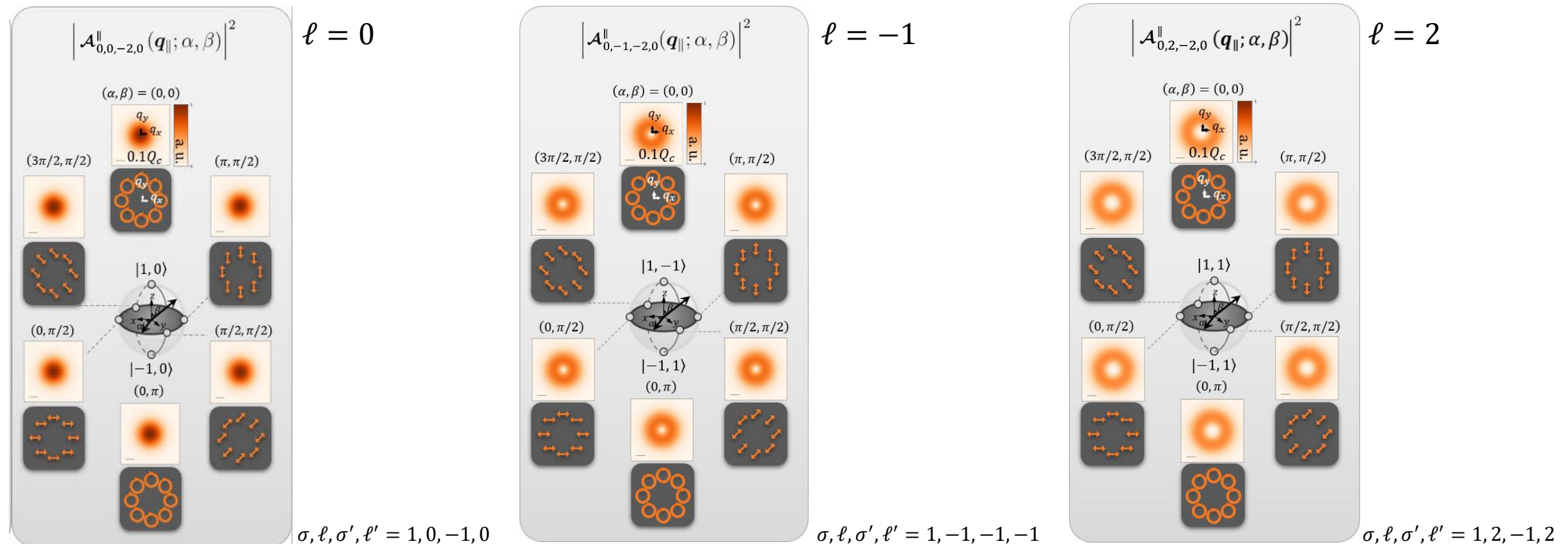
## 2. Transverse component : Vector vortex beams (VVB)

First case :  $\ell = \ell'$

$$\mathcal{A}_{0,\ell,-2,0}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = \tilde{F}_{|\ell|}(q_{\parallel}) e^{i(\ell\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \cos\left(\frac{\alpha}{2}\right) \hat{x} + \sin\left(\frac{\alpha}{2}\right) \hat{y} \right\}$$

Doughnut shape

Linear polarization changed with  $\alpha$



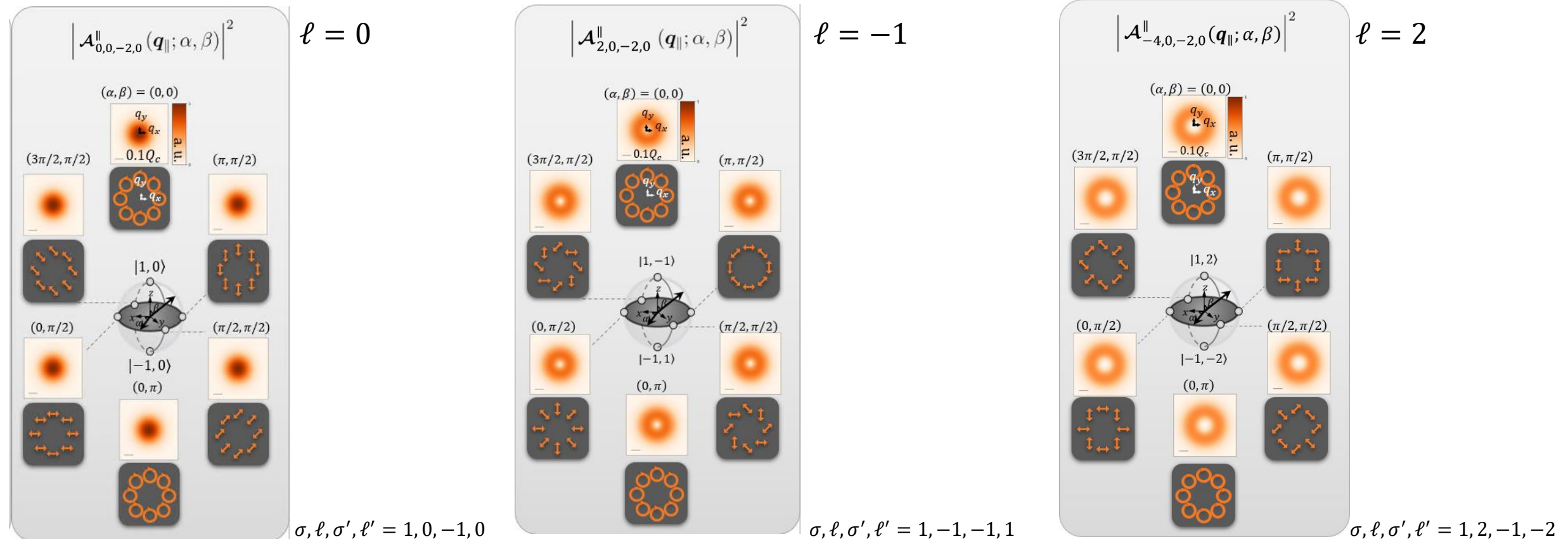
## 2. Transverse component : Vector vortex beams (VVB)

Second case :  $\ell = -\ell'$

$$\mathcal{A}_{-2\ell,0,-2,0}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\frac{\alpha}{2}} \left\{ \cos\left(-\ell\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{x} + \sin\left(-\ell\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{y} \right\}$$

Doughnut shape

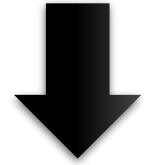
Linear polarization changed with  $\alpha$  and  $\phi_{q_{\parallel}}$



## 2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB:

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta\sigma, \bar{\sigma}}^z(\mathbf{q}_{\parallel}; \alpha, \beta) = -\frac{\sin \theta_{\mathbf{q}}}{\sqrt{2}} e^{i((\bar{\ell} + \bar{\sigma})\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} ((\Delta\ell + \Delta\sigma)\phi_{q_{\parallel}} + \alpha) \right] \left[ \cos \left( \frac{\beta}{2} \right) \tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) + \sin \left( \frac{\beta}{2} \right) \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel}) \right] \right. \\ \left. - i \sin \left[ \frac{1}{2} ((\Delta\ell + \Delta\sigma)\phi_{q_{\parallel}} + \alpha) \right] \left[ \cos \left( \frac{\beta}{2} \right) \tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) - \sin \left( \frac{\beta}{2} \right) \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel}) \right] \right\} \hat{\mathbf{z}}$$



Define  $\Delta J \equiv \Delta\ell + \Delta\sigma$     $\bar{J} \equiv \bar{\ell} + \bar{\sigma}$

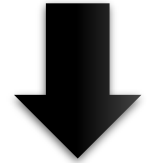
Longitudinal component of VVB:  $\beta = \frac{\pi}{2}$

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_{\mathbf{q}}}{2} e^{i(\bar{J}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} (\Delta J\phi_{q_{\parallel}} + \alpha) \right] [\tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) + \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel})] - i \sin \left[ \frac{1}{2} (\Delta J\phi_{q_{\parallel}} + \alpha) \right] [\tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) - \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel})] \right\} \hat{\mathbf{z}}$$

## 2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB:

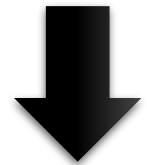
$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta\sigma, \bar{\sigma}}^z(\mathbf{q}_{\parallel}; \alpha, \beta) = -\frac{\sin \theta_{\mathbf{q}}}{\sqrt{2}} e^{i((\bar{\ell} + \bar{\sigma})\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} ((\Delta\ell + \Delta\sigma)\phi_{q_{\parallel}} + \alpha) \right] \left[ \cos \left( \frac{\beta}{2} \right) \tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) + \sin \left( \frac{\beta}{2} \right) \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel}) \right] \right. \\ \left. - i \sin \left[ \frac{1}{2} ((\Delta\ell + \Delta\sigma)\phi_{q_{\parallel}} + \alpha) \right] \left[ \cos \left( \frac{\beta}{2} \right) \tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) - \sin \left( \frac{\beta}{2} \right) \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel}) \right] \right\} \hat{\mathbf{z}}$$



Define  $\Delta J \equiv \Delta\ell + \Delta\sigma$   $\bar{J} \equiv \bar{\ell} + \bar{\sigma}$

Longitudinal component of VVB:  $\beta = \frac{\pi}{2}$

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_{\mathbf{q}}}{2} e^{i(\bar{J}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} (\Delta J\phi_{q_{\parallel}} + \alpha) \right] [\tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) + \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel})] - i \sin \left[ \frac{1}{2} (\Delta J\phi_{q_{\parallel}} + \alpha) \right] [\tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) - \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel})] \right\} \hat{\mathbf{z}}$$



If  $\tilde{F}_{|\bar{\ell} - \frac{\Delta\ell}{2}|}(q_{\parallel}) = \tilde{F}_{|\bar{\ell} + \frac{\Delta\ell}{2}|}(q_{\parallel})$

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_{\mathbf{q}}}{2} \tilde{F}_{|\bar{\ell} \pm \frac{\Delta\ell}{2}|}(q_{\parallel}) e^{i(\bar{J}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \cos \left[ \frac{1}{2} (\Delta J\phi_{q_{\parallel}} + \alpha) \right] \hat{\mathbf{z}}$$

**Doughnut shape**

**Azimuthal structure**

## 2. Longitudinal component : Vector vortex beams (VVB)

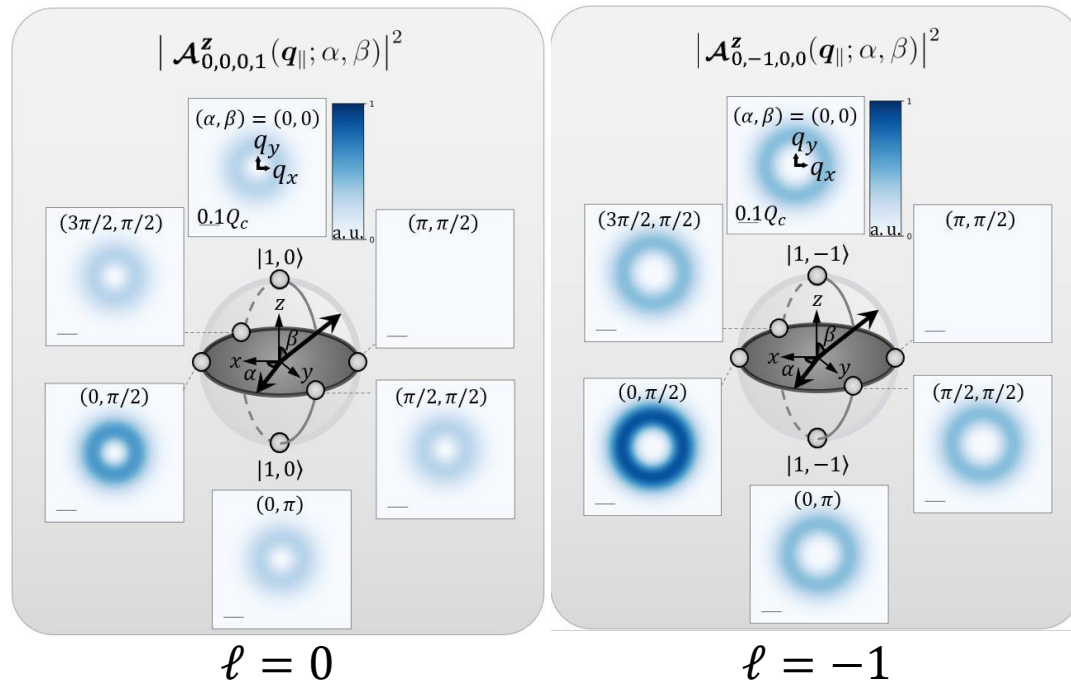
First case :  $\ell = \ell'$

$$\mathcal{A}_{0,\ell,\Delta\sigma,\ell+\bar{\sigma}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} \tilde{F}_{|\ell|}(q_{\parallel}) e^{i((\ell+\bar{\sigma})\phi_{q_{\parallel}} + \frac{\alpha}{2})} \cos \left[ \frac{1}{2} (\Delta\sigma \phi_{q_{\parallel}} + \alpha) \right] \hat{\mathbf{z}}$$

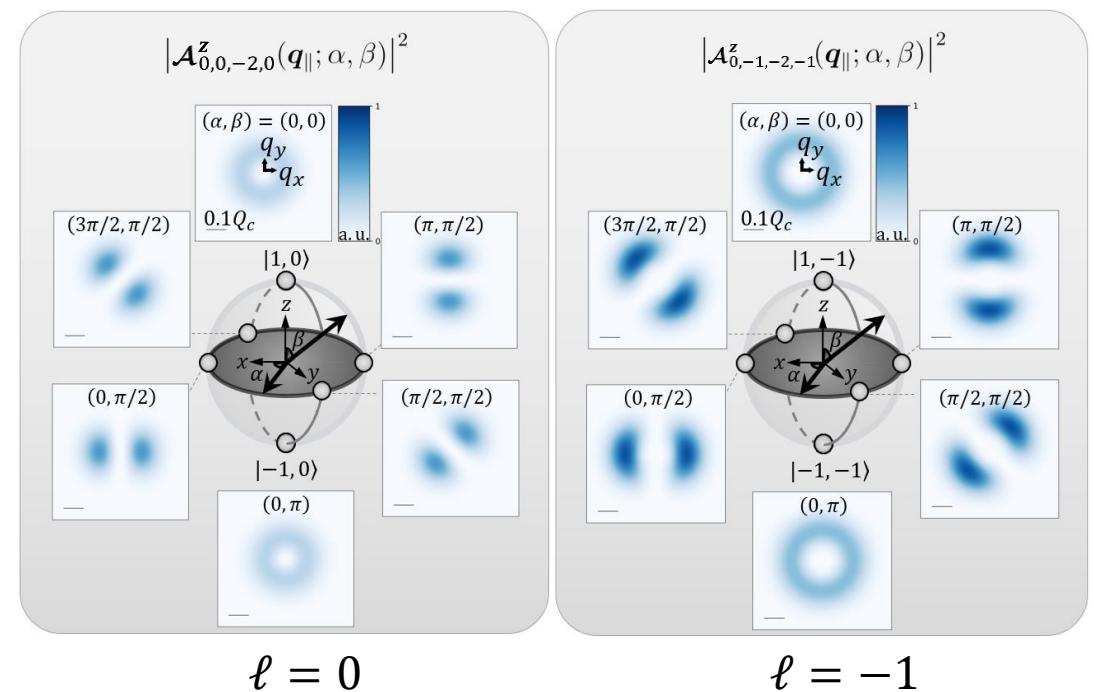
**Doughnut shape**

**Azimuthal structure**

Same spin :  $\Delta\sigma = 0 \Rightarrow 0 \text{ node}$



Opposite spin :  $\Delta\sigma = -2 \Rightarrow 2 \text{ nodes}$





## 2. Longitudinal component : Vector vortex beams (VVB)

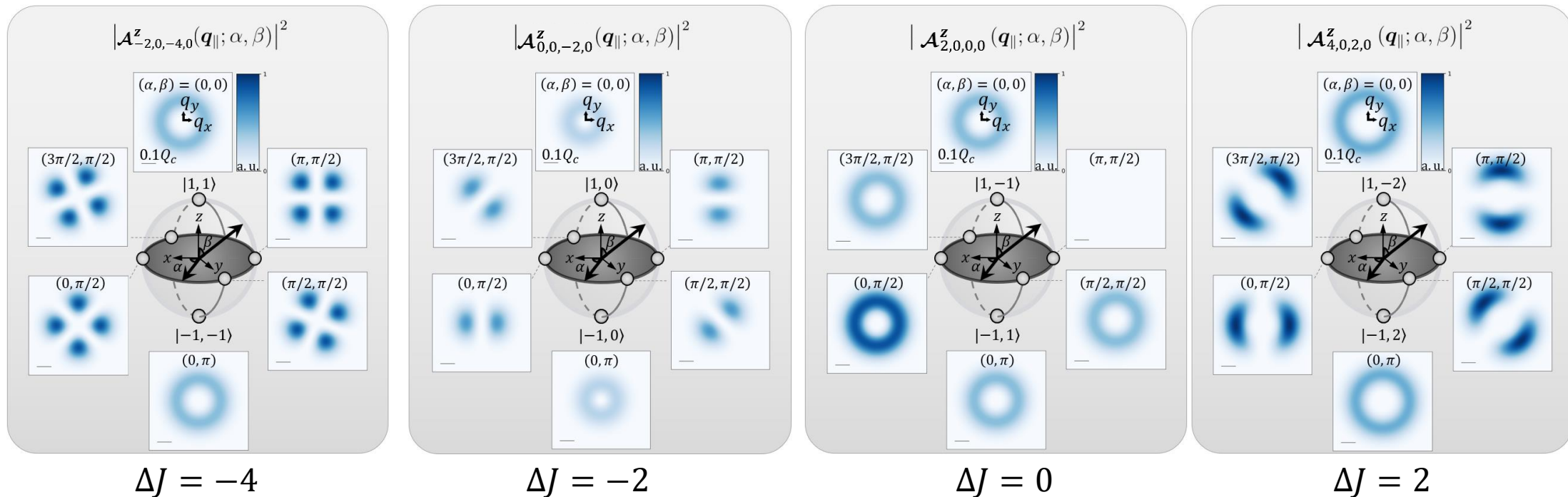
Second case :  $\ell = -\ell'$

$$\mathcal{A}_{-2\ell,0,-2\ell+\Delta\sigma,\bar{\sigma}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} \tilde{F}_{|\ell|}(q_{\parallel}) e^{i(\bar{\sigma}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \cos \left\{ \frac{1}{2} [(-2\ell + \Delta\sigma)\phi_{q_{\parallel}} + \alpha] \right\} \hat{z}$$

Doughnut shape

Azimuthal structure

Total AM difference:  $\Delta J = -2\ell + \Delta\sigma \Rightarrow$  **even number of nodes**



## 2. Longitudinal component : Vector vortex beams (VVB)

Can we have **odd** numbers of nodes ?

## 2. Longitudinal component : Vector vortex beams (VVB)

Can we have **odd** numbers of nodes ?

Longitudinal component of VVB:  $\beta = \frac{\pi}{2}$

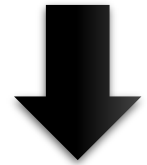
$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} (\Delta J \phi_{q_{\parallel}} + \alpha) \right] [\tilde{F}_{|\ell|}(q_{\parallel}) + \tilde{F}_{|\ell'|}(q_{\parallel})] - i \sin \left[ \frac{1}{2} (\Delta J \phi_{q_{\parallel}} + \alpha) \right] [\tilde{F}_{|\ell|}(q_{\parallel}) - \tilde{F}_{|\ell'|}(q_{\parallel})] \right\} \hat{\mathbf{z}}$$

## 2. Longitudinal component : Vector vortex beams (VVB)

Can we have **odd** numbers of nodes ?

Longitudinal component of VVB:  $\beta = \frac{\pi}{2}$

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} (\Delta J \phi_{q_{\parallel}} + \alpha) \right] [\tilde{F}_{|\ell|}(q_{\parallel}) + \tilde{F}_{|\ell'|}(q_{\parallel})] - i \sin \left[ \frac{1}{2} (\Delta J \phi_{q_{\parallel}} + \alpha) \right] [\tilde{F}_{|\ell|}(q_{\parallel}) - \tilde{F}_{|\ell'|}(q_{\parallel})] \right\} \hat{\mathbf{z}}$$



Substitute  $\tilde{F}_{|\ell|}(q_{\parallel}) = (-i)^{|\ell|} F_{|\ell|}(q_{\parallel})$ , and define  $|\bar{\ell}| = (|\ell'| + |\ell|)/2$ ,  $\Delta|\ell| = |\ell'| - |\ell|$

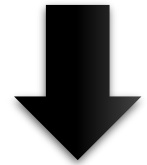
$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} - \frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} (\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta|\ell| + \alpha) \right] [F_{|\ell|}(q_{\parallel}) + F_{|\ell'|}(q_{\parallel})] + i \sin \left[ \frac{1}{2} (\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta|\ell| + \alpha) \right] [F_{|\ell|}(q_{\parallel}) - F_{|\ell'|}(q_{\parallel})] \right\} \hat{\mathbf{z}}$$

## 2. Longitudinal component : Vector vortex beams (VVB)

Can we have **odd** numbers of nodes ?

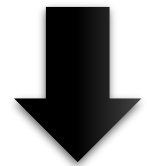
Longitudinal component of VVB:  $\beta = \frac{\pi}{2}$

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} (\Delta J \phi_{q_{\parallel}} + \alpha) \right] [\tilde{F}_{|\ell|}(q_{\parallel}) + \tilde{F}_{|\ell'|}(q_{\parallel})] - i \sin \left[ \frac{1}{2} (\Delta J \phi_{q_{\parallel}} + \alpha) \right] [\tilde{F}_{|\ell|}(q_{\parallel}) - \tilde{F}_{|\ell'|}(q_{\parallel})] \right\} \hat{\mathbf{z}}$$



Substitute  $\tilde{F}_{|\ell|}(q_{\parallel}) = (-i)^{|\ell|} F_{|\ell|}(q_{\parallel})$ , and define  $|\bar{\ell}| \equiv (|\ell'| + |\ell|)/2$ ,  $\Delta|\ell| \equiv |\ell'| - |\ell|$

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} - \frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} (\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta|\ell| + \alpha) \right] [F_{|\ell|}(q_{\parallel}) + F_{|\ell'|}(q_{\parallel})] - i \sin \left[ \frac{1}{2} (\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta|\ell| + \alpha) \right] [F_{|\ell|}(q_{\parallel}) - F_{|\ell'|}(q_{\parallel})] \right\} \hat{\mathbf{z}}$$



Define  $G_{|\ell|, |\ell'|}(q_{\parallel}) \equiv [F_{|\ell|}(q_{\parallel}) + F_{|\ell'|}(q_{\parallel})]$   $H_{|\ell|, |\ell'|}(q_{\parallel}) \equiv [F_{|\ell|}(q_{\parallel}) - F_{|\ell'|}(q_{\parallel})]$

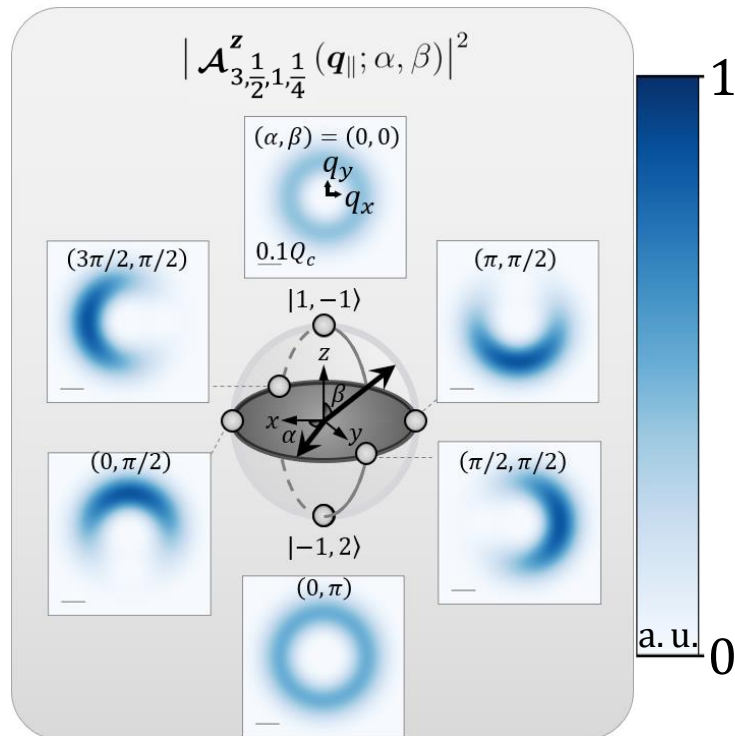
$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} - \frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} (\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta|\ell| + \alpha) \right] G_{|\ell|, |\ell'|}(q_{\parallel}) - i \sin \left[ \frac{1}{2} (\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta|\ell| + \alpha) \right] H_{|\ell|, |\ell'|}(q_{\parallel}) \right\} \hat{\mathbf{z}}$$

## 2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB:  $\beta = \frac{\pi}{2}$

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} - \frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} \left( \Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] G_{|\ell|, |\ell'|}(q_{\parallel}) - i \sin \left[ \frac{1}{2} \left( \Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] H_{|\ell|, |\ell'|}(q_{\parallel}) \right\} \hat{\mathbf{z}}$$

Example for odd number  $\Delta J : \sigma, \ell, \sigma', \ell' = 1, -1, -1, 2$



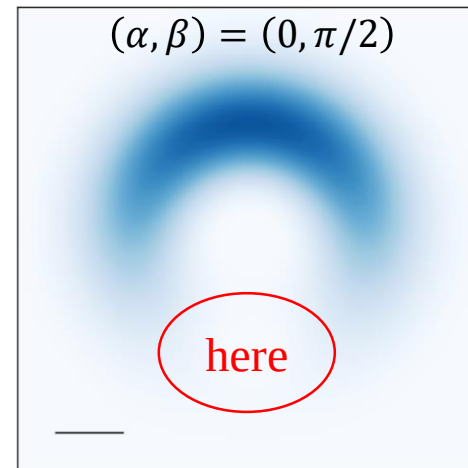
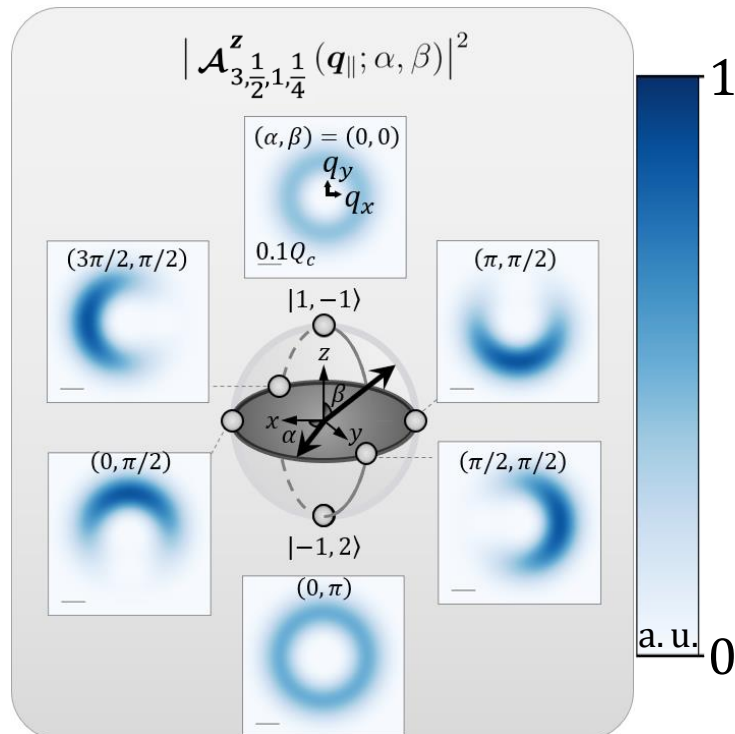
$\Delta J = 1 \Rightarrow \text{one node}$

## 2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB:  $\beta = \frac{\pi}{2}$

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} - \frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} \left( \Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] G_{|\ell|, |\ell'|}(q_{\parallel}) - i \sin \left[ \frac{1}{2} \left( \Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] H_{|\ell|, |\ell'|}(q_{\parallel}) \right\} \hat{z}$$

For example :  $\sigma, \ell, \sigma', \ell' = 1, -1, -1, 2$



**But !** For  $\alpha = 0$ , node appear when  $\phi_{q_{\parallel}} = \frac{3\pi}{2}$

$$\cos \left[ \frac{1}{2} \left( \Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] = \cos \left[ \frac{\pi}{2} \right] = 0$$

$$\sin \left[ \frac{1}{2} \left( \Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] = \sin \left[ \frac{\pi}{2} \right] = \mathbf{1}$$

**Maximum value !?**

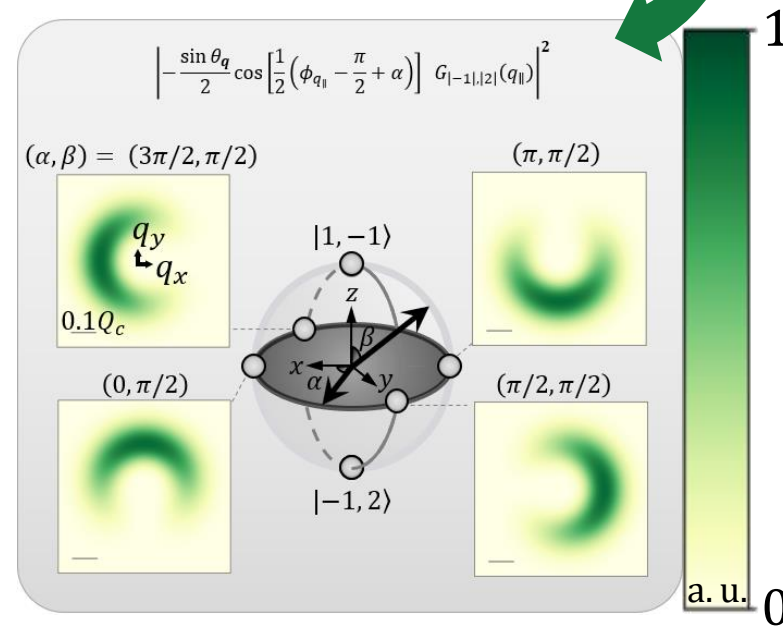
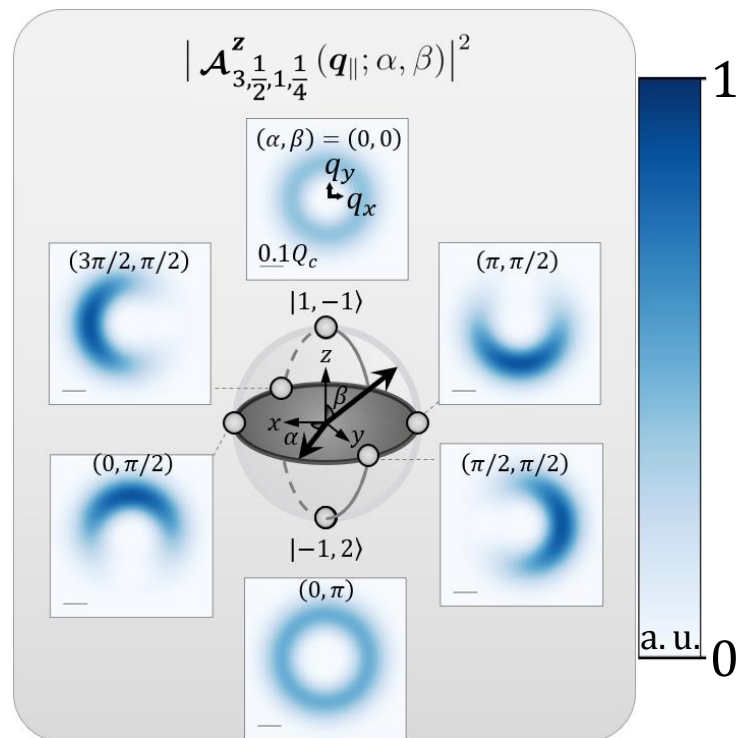


## 2. Longitudinal component : Vector vortex beams (VVB)

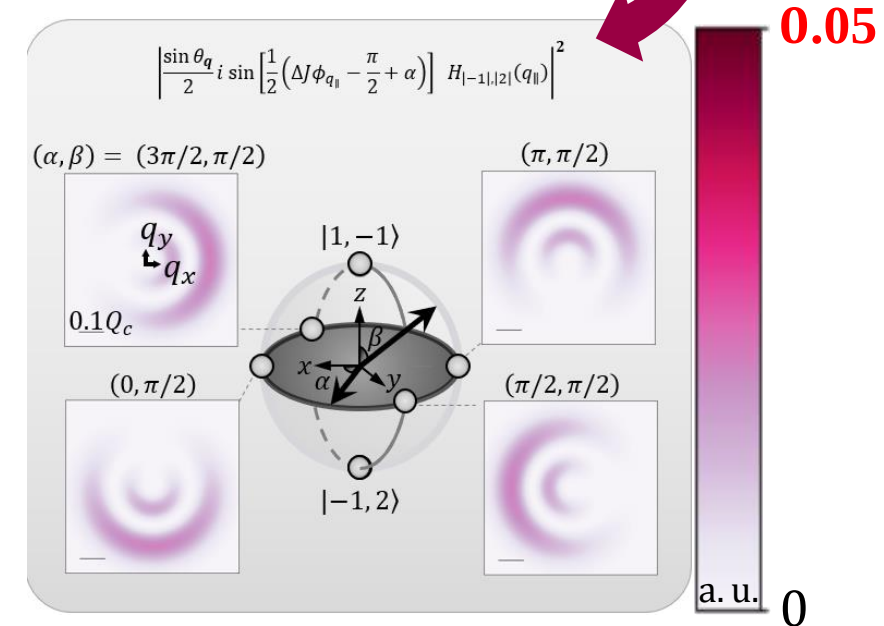
Longitudinal component of VVB:  $\beta = \frac{\pi}{2}$

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} - \frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} \left( \Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] G_{|\ell|, |\ell'|}(q_{\parallel}) - i \sin \left[ \frac{1}{2} \left( \Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] H_{|\ell|, |\ell'|}(q_{\parallel}) \right\} \hat{z}$$

For example :  $\sigma, \ell, \sigma', \ell' = 1, -1, -1, 2$



$$G_{|\ell|, |\ell'|}(q_{\parallel}) \equiv [F_{|\ell|}(q_{\parallel}) + F_{|\ell'|}(q_{\parallel})]$$



$$H_{|\ell|, |\ell'|}(q_{\parallel}) \equiv [F_{|\ell|}(q_{\parallel}) - F_{|\ell'|}(q_{\parallel})]$$

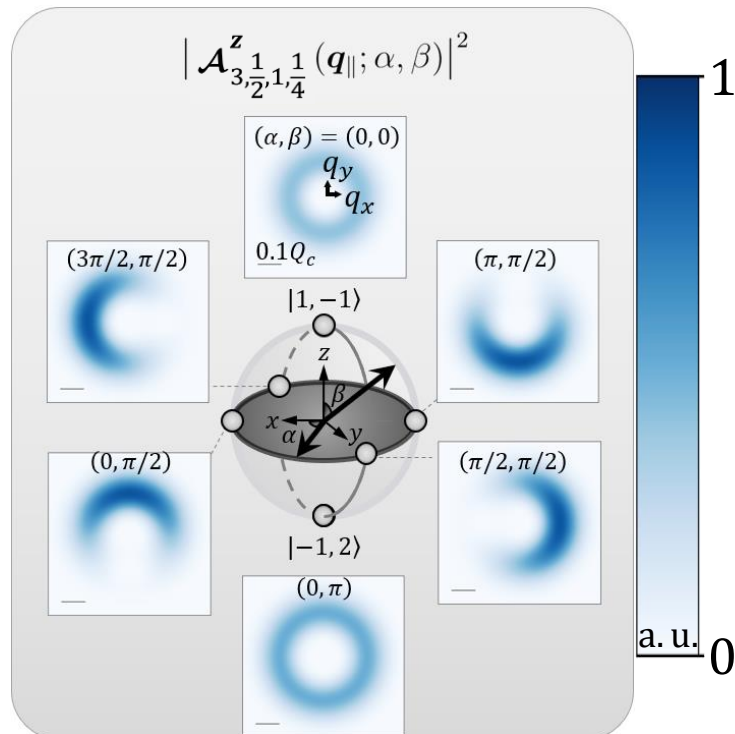
**Nearly zero**

## 2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB:  $\beta = \frac{\pi}{2}$

$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} - \frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} \left( \Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] G_{|\ell|, |\ell'|}(q_{\parallel}) - i \sin \left[ \frac{1}{2} \left( \Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] H_{|\ell|, |\ell'|}(q_{\parallel}) \right\} \hat{\mathbf{z}}$$

For example :  $\sigma, \ell, \sigma', \ell' = 1, -1, -1, 2$



What does it mean to have a horseshoe shape light?



## 2. Longitudinal component : Vector vortex beams (VVB)

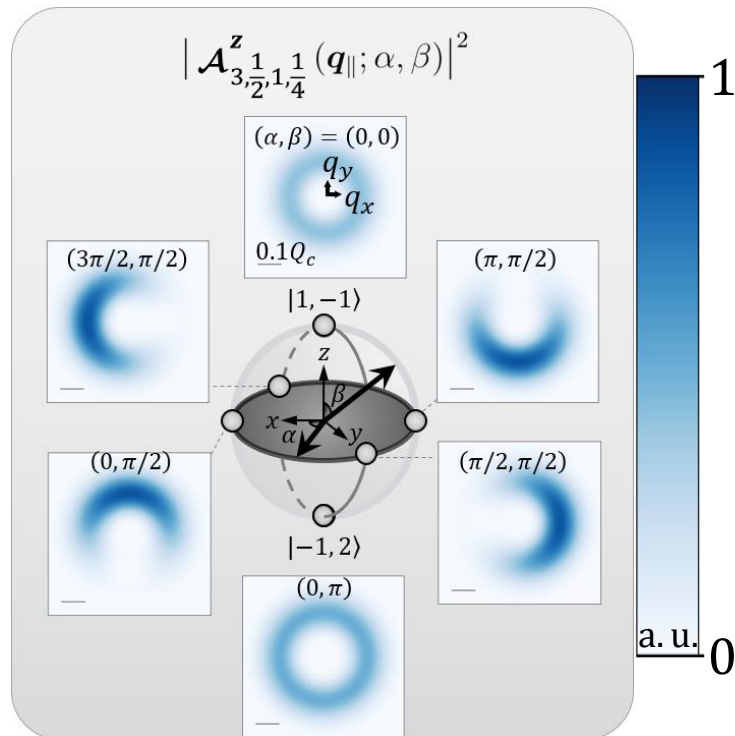
Longitudinal component of VVB:  $\beta = \frac{\pi}{2}$

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For example :  $\sigma, \ell, \sigma', \ell' = 1, -1, -1, 2$

**Find the real field !**

$$\text{Re} \left[ \mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) e^{-i\omega t} \right]$$



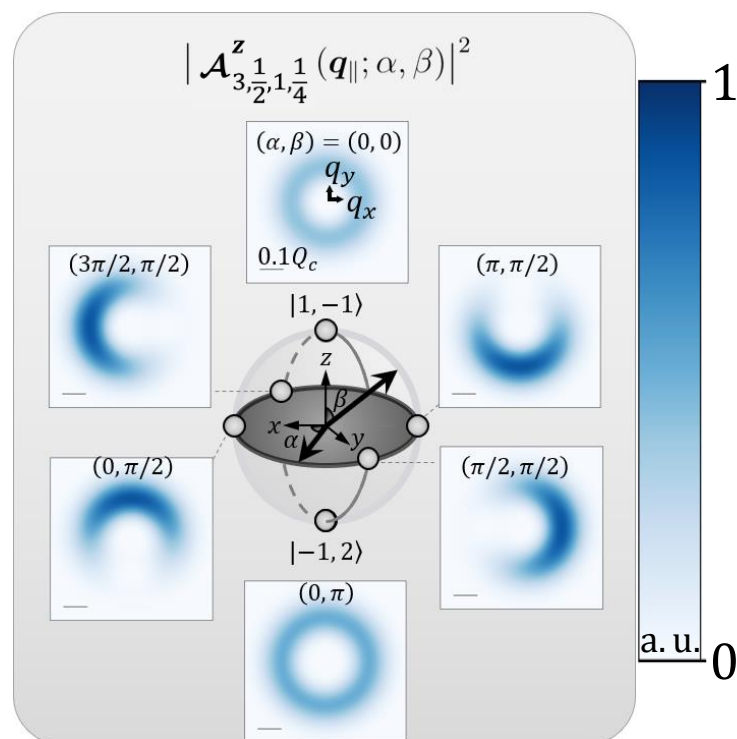
## 2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB:  $\beta = \frac{\pi}{2}$

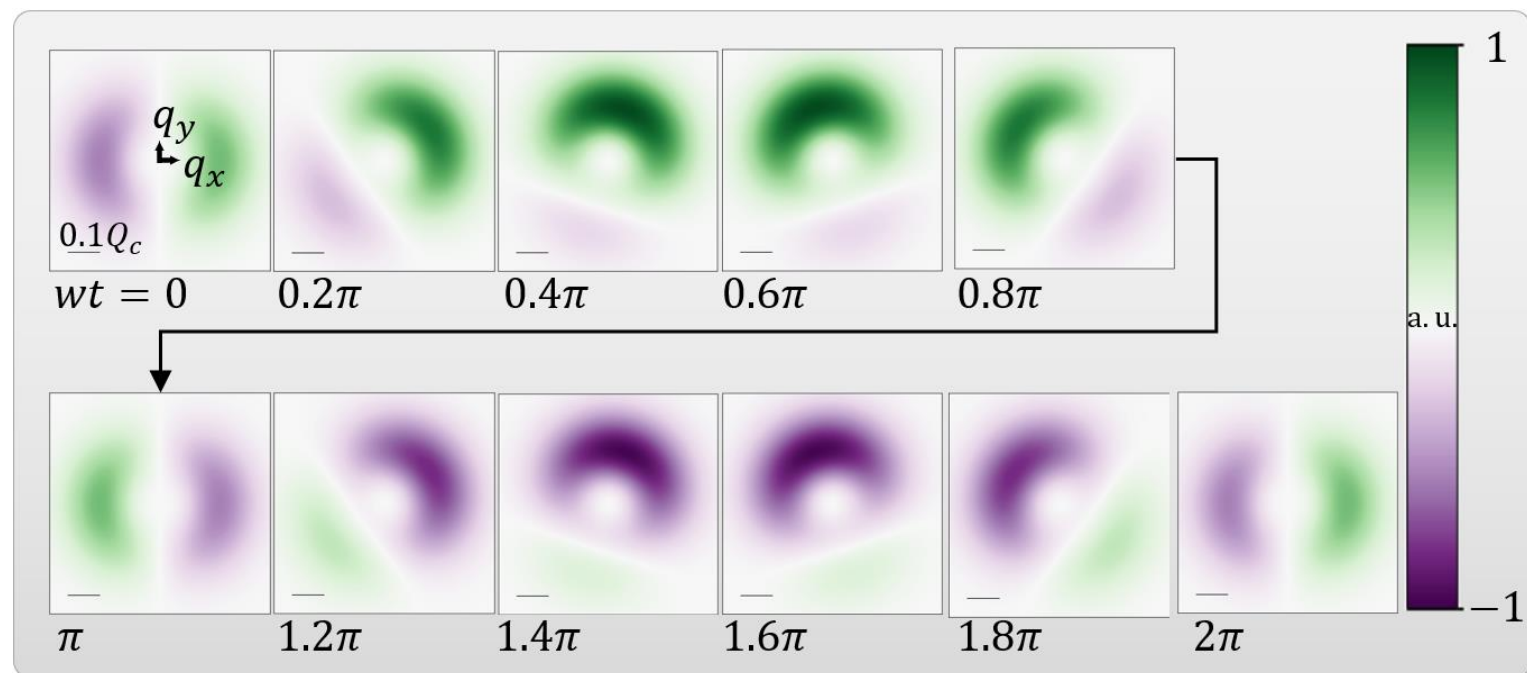
$$\mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} - \frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2})} \left\{ \cos \left[ \frac{1}{2} \left( \Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] G_{|\ell|, |\ell'|}(q_{\parallel}) - i \sin \left[ \frac{1}{2} \left( \Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] H_{|\ell|, |\ell'|}(q_{\parallel}) \right\} \hat{z}$$

For example :  $\sigma, \ell, \sigma', \ell' = 1, -1, -1, 2$

**Find the real field !**



$$\text{Re} \left[ \mathcal{A}_{\Delta\ell, \bar{\ell}, \Delta J, \bar{J}}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) e^{-i\omega t} \right]$$



1. Introduction

2. Spin-orbital angular momentum interaction in light

3. Light-matter interaction

- Encoding SOI information
- Selective excitation

4. Exciton wave packet

Summary and outlook

# 3. Light-matter interaction

1. Introduction
2. Spin-orbital angular momentum interaction in light
3. Light-matter interaction
- 4. Exciton wave packet**

Summary and outlook



# 4. Exciton wave packet

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